



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS**

**A MAJOR PROJECT REPORT
ON
CIRCUITRON: CIRCUIT RECOGNITION, TRANSFORMATION, AND ANALYSIS**

Submitted By:

Anveshan Timsina	(THA078BEI005)
Sandesh Dhital	(THA078BEI034)
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Submitted To:

Department of Electronics and Computer Engineering
Thapathali Campus
Kathmandu, Nepal

March 2026



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Submitted To:

Department of Electronics and Computer Engineering
Thapathali Campus
Kathmandu, Nepal

In partial fulfillment for the award of the
Bachelors Degree in Electronics, Communication and Information Engineering

Under the Supervision of

Er. Kobid Karkee

March 2026

DECLARATION

We hereby declare that the project entitled, “**CIRCUITRON: Circuit Recognition, Transformation, and Analysis**” in the partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering **Electronics, Communication and Information Engineering** is a bona fide report of the work carried out by us. These materials contained within the report have not been submitted to any University or Institution for the award of any degree, and we are the only author of this complete work and no sources other than the listed ones have been used in this work.

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CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the **Department of Electronics and Computer Engineering, IOE, Thapathali Campus**, a major project work entitled “**CIRCUITRON: Circuit Recognition, Transformation, and Analysis**” submitted by **Anveshan Timsina, Sandesh Dhital, Saroj Nagarkoti , Utsab Dahal** in partial fulfillment for the award of Bachelor of Engineering in Electronics, Communication and Information Engineering. The project was carried out under the special supervision and within the time prescribed by the syllabus.

We found that the students to be hardworking, skilled and ready to undertake any related work to their field of study and hence we recommend the award of partial fulfillment of Bachelor’s Degree of Electronics, Communication, and Information Engineering.

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ACKNOWLEDGEMENT

First, we would like to thank the Department of Electronics and Computer Engineering for providing us with an opportunity to learn and research on this project. We appreciate the department's assistance and recommendations in the selection and hopefully the successful completion of this project.

Special thanks to our supervisor, **Er. Kovid Karkee**, for his guidance, invaluable feedback, and constant support in regards to the project. His expertise and encouragement have been pivotal in shaping the direction, scope and implementation of this project.

Additionally, we would like to thank all of our teachers, friends, and other direct as well as indirect contributors for their indispensable support and encouragement during our study and execution of the project. We are also grateful to the authors of various papers that we have referenced for our project.

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ABSTRACT

CIRCUITRON is an integrated system designed to bridge the gap between traditional hand-drawn electrical schematics and modern circuit simulation tools. The project pipeline automates the transformation of handwritten circuit diagrams into CircuitJS-compatible text files which enables the users to interact with, simulate, and analyze circuits in an editable digital environment. Deep learning models such as YOLOv7 for component detection, two kinds of OCR models, TrOCR and also a custom OCR pipeline for text recognition, work together with BFS-based path finding wire detection algorithm to achieve the project's objectives of visualizing circuits as editable schematics, and allowing simulation. The system has achieved a mAP@0.5 of 95% for component detection, and a character accuracy of 84.5% for text recognition, and proper wire and node detection, for horizontal, vertical as well as diagonal wires, across varied drawing styles. A user-friendly interface allows manual corrections, schematic editing, and direct integration with CircuitJS for simulation. Additionally, the platform incorporates a DeepSeek v3.1-powered assistant to answer circuit-related queries, thus reinforcing learning through natural language interaction.

Keywords: circuit analysis, Multi-head BFS, simulation, TrOCR, YOLOv7.

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LIST OF ABBREVIATIONS

AC	Alternating Current
AI	Artificial Intelligence
AP	Average Precision
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BJT	Bipolar Junction Transistor
CER	Character Error Rate
CGHD	Circuit Graph Handwritten Diagrams
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CTC	Connectionist Temporal Classification
DC	Direct Current
DL	Deep Learning
DOM	Document Object Model
EDA	Electronic Design Automation
ELAN	Efficient Layer Aggregation Network
FET	Field-Effect Transistor
FSM	Finite State Machine
GNN	Graph Neural Network
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HCD	Handwritten Circuit Diagram
HSV	Hue, Saturation, Value
HTTP	Hypertext Transfer Protocol
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
IoU	Intersection over Union
JSON	JavaScript Object Notation
LLM	Large Language Model
LMDB	Lightning Memory-Mapped Database
LSTM	Long Short-Term Memory
LTspice	Linear Technology Simulation Program with Integrated Circuit Emphasis
mAP	Mean Average Precision
ML	Machine Learning
NLP	Natural Language Processing
NMS	Non-Maximum Suppression

OCR	Optical Character Recognition
PAN	Path Aggregation Network
PCB	Printed Circuit Board
PHT	Probabilistic Hough Transform
Python	high-level general-purpose programming language
REST	Representational State Transfer
RF	Radio Frequency
RMS	Root Mean Square
SGD	Stochastic Gradient Descent
SPICE	Simulation Program with Integrated Circuit Emphasis
VQA	Visual Question Answering
YOLO	You Only Look Once

1 INTRODUCTION

“CIRCUITRON: Circuit Recognition, Transformation, and Analysis” is a system to convert hand-drawn electrical circuit diagrams into CircuitJS-compatible text files enabling simulation and graphical analysis along with digital visualization of the circuit in an editable schematic. The system also integrates a prompt-based Artificial Intelligence (AI) assistant (DeepSeek v3.1) to help answer queries related to the relevant circuit.

1.1 Background Information

The first step of learning about electrical and electronic circuits involves looking at and hand-drawing said circuits with the necessary components. These hand-drawn circuit diagrams continue to play an important role in electronics education and early-stage hardware design due to the convenience they offer. However, such hand-drawn diagrams prove to be limited in their utility in the current education landscape that prioritizes intuitive, visual understanding as they require significant mental gymnastics to facilitate any form of analysis of the circuits, more so when incorporating variation in parameters.

A slew of rather easy-to-use circuit simulation and analysis tools such as Linear Technology Simulation Program with Integrated Circuit Emphasis (LTspice) and KiCad exist, of course; and they continue to foster visual learning and analysis of circuits of a range of complexities. But these traditional Electronic Design Automation (EDA) tools lack the capability to accept hand-sketched circuit diagrams as input. Instead, users are required to redraw the circuits in software using Graphical User Interface (GUI) tools or by manually entering Simulation Program with Integrated Circuit Emphasis (SPICE) netlists; steps that are both tedious and error-prone [1, 2]. Be it due to the (admittedly not very steep yet inconvenient) learning curve of these tools or just the additional effort of having to redraw a circuit digitally (the dullness of which only goes up with circuit complexity), the instructors as well as the students often tend to neglect arguably essential simulation-based intuitive teaching and learning. This again encourages the same long-established rote learning practice instead of ushering in a more modern way of understanding circuits and electronics at large. It has been long proven that students tend to perform better when given the appropriate simulative manipulation and concept clarification tools, especially in the context of electrical and electronics circuits [3]. It then follows that the reluctance to use such tools is actively hindering the students’ potential. Thus, the use of visual manipulative simulation tools is the need of the hour. In fact, it has been the need of the hour for quite a while. But, due to a lack of any uber-convenient method, it is yet to be popularly embraced, primarily in technologically nascent countries such as Nepal.

1.2 Motivation

In recent years, the mantra of the modern world has seemingly become “adapt or die”. The society regularly stomps on the tech-illiterate to make way for the ones willing to champion the use of new and shiny tools in all aspects of their lives. In an increasingly globalized economy, this is especially true for the up-and-coming students who need to compete with other students all over the world relying solely on their understanding of subject matter in their particular fields. In such case, it becomes essential that they are given the tools to enhance their learning and understanding even by minor increments. In the context of electronics, circuits are the fundamental building blocks that permeate each and every aspect of our “teched-out” world and it is important for learners in the field to have an intuitively in-depth understanding of this fundamental ingredient. Thus, there exists a motivation for a hassle-free method to analyze and understand circuits to deepen the necessary conception and comprehension.

The motivation stems not only from a deep-seated commitment to give the instructors and students tools to easily visualize and analyze circuits in the belief that it will genuinely improve the quality of professionals and engineers born out of universities, but also from the personal experience of the members involved in this project. As the electronics engineers of the near future who have had and will continue to have to deal with circuits in one form or another, the members involved in this project are uniquely positioned to understand the plight of a beginner in the study of electronics. And a tool which can convert a hand-drawn circuit into a digital schematic and also allow for analysis would go a long way in alleviating that plight.

1.3 Problem Statement

The EDA tools currently available do not support the direct transformation of hand-drawn circuit images into simulation-ready schematics. There exist a few nascent prototypes for the same but there seems to be no widely adopted, robust, and user-friendly pipeline that combines component recognition, reliable wire connectivity, schematic generation and simulation, whether by itself or by integration with SPICE simulators.

“CIRCUITRON: Circuit Recognition, Transformation, and Analysis” project aims to create an integrated system capable of first, recognizing and digitizing hand-drawn circuits into editable schematics and then allowing the user to analyze those circuits through graphs using CircuitJS-compatible text files with an added AI assistant to handle related queries.

1.4 Project Objectives

- To automatically convert hand-drawn circuit diagrams into CircuitJS-compatible text files for simulation.
- To visualize the circuit in an editable schematic in addition to an AI assistant for handling simple, relevant queries.

1.5 Project Scope and Applications

1.5.1 Project Scope

This project focuses on the development of a software pipeline that accepts scanned or photographed images of hand-drawn electrical circuits and transforms them into CircuitJS-compatible netlists for simulation as well as editable schematics through an intermediate integrated data structure to facilitate the transformation. The scope includes detecting common circuit components (14, to be exact, including supplementary components like gnd, crossover, etc.) and their values in mentioned units. It infers wire connectivity with junction recognition and generates a circuit diagram with simulation capabilities. Additionally, the system incorporates an Large Language Model (LLM)-based (Deepseek v3.1) query-solving assistant. The system excludes any form of advanced circuits such as those associated with Radio Frequency (RF) or mixed-signal, and Printed Circuit Board (PCB) designs.

1.5.2 Project Applications

- The system is foremost, an educational tool designed to assist students and instructors alike in conveniently digitizing circuit diagrams and understanding circuit behavior through limited simulation and interactive Q&A.
- The project enables quick feedback and validation of hand-drawn designs against simulation results early in the design process.
- The project can facilitate quick on-the-go analysis of circuits for students and potentially, field engineers as well.

1.5.3 Project Limitations

Despite its innovative concept, the system is constrained by several factors. First, its detection and recognition accuracy is heavily dependent on the clarity of input images. Low-res or messy sketches may lead to recognition errors. Moreover, the system supports only a fixed set of commonly used components. This makes it unable to handle rare components and limits its functionality. Wire connectivity mapping is achieved through a junction-based approach, which assumes standard drawing practices. However,

there is no one way to draw a circuit diagram; non-standard and overlapping wires, or complicated circuits may confuse the system. Additionally, the system lacks a simulation engine of its own and instead uses an open-source tool like CircuitJS. This limits the simulation capabilities to that of CircuitJS. More rigorous training on appropriate data would help remedy some of these issues, however finding the ideal dataset with enough handwritten examples for all kinds of components and circuits has proven challenging.

1.6 Report Organization

The report begins with an Introduction which includes background information, motivation, problem statement, project objectives, applications, and limitations. Literature Review presents a summary of existing relevant research and their corresponding limitations that are in turn addressed by this project. Following this, Requirement Analysis simply describes the project requirements i.e, software needs, on account of it being a software-only project, along with feasibility study (economic, time, operational, technical). System Architecture and Methodology explains the system design through diagrams like block diagram, flowchart, use-case diagram, and architecture diagrams for specific modules used in the project. These diagrams are supplemented with serviceable explanations for each element in the project pipeline. It also includes dataset analysis and performance metrics expectation tables. Implementation Details clarifies practical aspects of the system, starting with preprocessing techniques, and then the model training details. This is followed by relevant algorithms for wire detection and connectivity and error handling techniques. Results and Analysis detail project outcomes i.e, the results of object detection, text recognition and line detection, in addition to performance analysis of different models used (with the use of graphs). Future Enhancements propose potential improvements or extensions and the Conclusion presents a summary of the entire project from problem statement to results and recommendations for future work. Appendices contain budget breakdown and Gantt chart as supplementary materials supporting the main content of the report.

2 LITERATURE REVIEW

The automatic recognition of hand-drawn circuit diagrams has been explored since the early 2000s. In one of the earliest studies, Edwards and Chandran [4] proposed an approach that first identified circuit components and then inferred node segments by considering spatial relationships between them. Their system achieved an accuracy of 86% for component recognition and 92.5% for node identification, with wire connections traced between identified nodes using hard-coded heuristic conditions. While pioneering for its time, this approach was inherently brittle due to its reliance on manually crafted rules and could not generalize to diverse handwriting styles or circuit topologies. Ramel et al. [5] addressed a related problem in the broader domain of technical document recognition, proposing a structured approach that separated graphical primitives from textual annotations followed by structural matching against predefined symbol templates. Belongie et al. [6] introduced shape context descriptors for object recognition, computing histograms of relative contour point positions to provide descriptors invariant to small deformations, a useful property for hand-drawn symbols. While these early methods laid the groundwork, they were limited by their inability to learn discriminative features from data, motivating the shift toward deep learning-based approaches.

The detection and classification of individual circuit components is a foundational step in any circuit recognition pipeline, and modern approaches rely heavily on deep learning-based object detection architectures. The R-Convolutional Neural Network (CNN) family of detectors introduced region proposals for object detection: Girshick et al. [7] proposed R-CNN using selective search for candidate regions, followed by Fast R-CNN [8] which shared convolutional features across proposals, and Faster R-CNN [9] which replaced selective search with a Region Proposal Network. He et al. [10] extended this with Mask R-CNN, adding instance segmentation capabilities. In the circuit domain, Bayer et al. [11] leveraged Mask R-CNN for instance segmentation of components in hand-drawn diagrams, achieving approximately 94% mask accuracy, although they noted that symbol recognition was “reasonable, yet incomplete,” particularly in complex drawings with inconsistent handwriting. Single-shot detectors eliminate the region proposal stage for real-time inference; Liu et al. [12] introduced SSD, which discretizes the output space into default bounding boxes at multiple scales. The YOLO family has been especially influential: Redmon et al. [13] proposed the original YOLO architecture framing detection as a single regression problem, followed by YOLOv2 [14] with batch normalization and anchor boxes, YOLOv3 [15] with feature pyramid predictions at three scales, YOLOv4 [16] with Mish activation and Cross-Stage Partial connections, and YOLOv5 [17] which gained widespread adoption through its PyTorch implementation. Wang et al. [18] proposed YOLOv7, introducing the (E-Efficient Layer Aggregation Network (ELAN)) and model re-parameterization techniques that achieved

state-of-the-art results on MS COCO, outperforming all prior real-time detectors in both speed and accuracy. YOLOv7 is the architecture employed in the present project for component detection.

In a 2021 study, Adnan et al. [19] developed a two-stage CNN-based system for classifying 20 different hand-drawn circuit components, achieving a recognition accuracy of 97.33%. Reddy and Panicker [1] combined YOLOv5 for component detection with probabilistic Hough transform-based node recognition, achieving a mAP of 98.2% in component detection and an overall schematic generation accuracy of 80%, though their system did not handle labeled components. Abhinav et al. [20] proposed a two-step method of detecting and classifying components followed by feeding labeled component strings into an Finite State Machine (FSM) for circuit recognition.

Extracting component values such as “10k Ω ” or “5V” from hand-drawn annotations is critical for generating functionally accurate netlists. Shi et al. [21] proposed the Convolutional Recurrent Neural Network (CRNN) architecture, integrating convolutional feature extraction, bidirectional Long Short-Term Memory (LSTM) sequence modeling, and Connectionist Temporal Classification (CTC)-based transcription into a unified framework that has become the de facto standard for scene text recognition and forms the basis of the OCR module in the present project. Graves et al. [22] originally introduced the CTC loss function for labeling unsegmented sequence data, with the key advantage that it does not require pre-segmented training data, which is critical for handwritten text where character boundaries are ambiguous. Baek et al. [23] conducted a comprehensive comparison of scene text recognition models, revealing that attention-based decoders generally outperform CTC-based decoders on irregular text while CTC remains competitive on regular text, informing the design decision in this project to adopt a CTC-based CRNN for circuit text which tends to be horizontally aligned. More recently, transformer-based architectures such as TrOCR [24] have demonstrated strong performance on handwritten text recognition by leveraging pre-trained vision transformers, though they require substantially more computational resources, making them less suitable for lightweight systems like the one proposed here.

Detecting wires and inferring electrical connectivity is arguably the most challenging aspect of circuit diagram recognition, as wire detection must contend with arbitrary orientations, intersections, crossovers, and junctions. The Hough transform, originally proposed by Duda and Hart [25], is a classical technique for detecting straight lines, and its probabilistic variant introduced by Matas et al. [26] operates more efficiently on random subsets of edge points. Reddy and Panicker [1] used the Probabilistic Hough Transform for wire detection in hand-drawn circuits but encountered difficulties with curved or overlapping wires. Morphological operations such as the parallel thinning

algorithm proposed by Zhang and Suen [27] have also been employed for wire extraction, reducing binary regions to single-pixel-wide skeletons while preserving topological connectivity. Hemker et al. [28] proposed a three-stream approach (one for component detection, one for line and junction detection, and one for text recognition) with outputs combined into a JavaScript Object Notation (JSON) representation from which a netlist is derived. Kelly and Cole [29] addressed wire detection in printed circuit schematics, achieving 86.4% digitization success and 99.6% OCR accuracy, though their system was trained exclusively on computer-generated diagrams and cannot handle hand-drawn variability. An alternative approach formulates wire tracing as a graph search problem where the binarized circuit image is treated as a grid graph and breadth-first search or A^* algorithms find paths between component terminals; the present project adopts this path-finding approach after initial experiments with Hough transform-based methods proved inadequate.

Representing circuits as graphs, where nodes correspond to components or connection points and edges represent electrical connections, enables topological reasoning and downstream simulation. Kipf and Welling [30] introduced Graph Convolutional Networks that generalize convolutions to graph-structured data. Yamakaji et al. [31] developed Circuit2Graph, transforming circuit diagrams into graph structures and achieving 97.9% accuracy on circuit classification tasks using Graph Neural Networks (GNNs), demonstrating that graph neural networks can effectively model the global structure of a circuit even when visual features alone are insufficient.

The ultimate goal of many circuit recognition systems is to produce a netlist that can drive simulation. Traditional EDA tools such as KiCad and LTspice support netlist export from manually created schematics but do not allow direct image-to-netlist conversion [32]. Sertdemir et al. [2] developed Img2Sim, an artificial neural network-based system that detects components, infers connectivity, and outputs a complete SPICE netlist with over 90% accuracy in topology extraction, though it requires clean input images and is limited to a predefined set of components. Nagel and Pederson [33] developed SPICE at the University of California, Berkeley, establishing the standard for circuit simulation that persists to this day, with modern derivatives including Ngspice for Direct Current (DC), Alternating Current (AC), and transient analysis, and commercial tools such as PSpice and Simulink. CircuitJS [34], a web-based interactive circuit simulator providing real-time visualization of voltage, current, and waveform behavior, is the simulation platform adopted by the present project due to its open format and accessibility.

The emergence of LLMs has opened new possibilities for circuit understanding and explanation. The CircuitVQA dataset contains more than 115,000 question-image pairs based on 5,725 unique circuits covering approximately 70 symbol types [35], supporting

transformer-based Visual Question Answering (VQA) models that outperform generic systems when fine-tuned on domain-specific data. Vungarala et al. [36] developed SPICEPilot, leveraging GPT-based models to generate and interpret SPICE code, though general-purpose models often fail to respect domain conventions, underscoring the need for domain-adapted models. The present project integrates DeepSeek as an LLM-based assistant for answering circuit-related queries. Chen et al. [3] studied the efficacy of simulation-based learning in electronics education, finding that interactive circuit simulations significantly improved student comprehension, motivating the design of CIRCUITRON as not merely a recognition tool but an educational platform.

While substantial progress has been made in component recognition, value extraction, connectivity inference, netlist generation, and interactive Q&A, existing systems remain fragmented or limited in scope. Most systems address only one or two stages of the recognition pipeline; several [28, 29] were designed for printed schematics and do not generalize to hand-drawn circuits; none generate an editable, interactive schematic; VQA and LLM-based tools have not been integrated into recognition pipelines; and many systems [1, 11] do not extract handwritten component values. CIRCUITRON addresses these gaps by providing a unified, end-to-end pipeline that takes a hand-drawn circuit image as input and produces an editable, simulation-ready digital schematic with integrated LLM-based assistance. Table 2-1 summarizes the capabilities of the reviewed systems.

Table 2-1 Comparison of Existing Circuit Recognition Systems

System	Hand-drawn	Component Detection	Value Extraction	Wire Detection	Netlist/Simulation	AI Assist
Edwards [4]	Yes	Yes	No	Yes	No	No
Adnan et al. [19]	Yes	Yes	No	No	No	No
Kelly & Cole [29]	No	Yes	Yes	Yes	No	No
Reddy & Panicker [1]	Yes	Yes	No	Yes	Yes	No
Bayer et al. [11]	Yes	Yes	No	Yes	No	No
Hemker et al. [28]	No	Yes	Yes	Yes	Yes	No
Img2Sim [2]	Yes	Yes	No	Yes	Yes	No
Circuit2Graph [31]	No	Yes	No	No	No	No
SPICEPilot [36]	No	No	No	No	Yes	Yes
CIRCUITRON	Yes	Yes	Yes	Yes	Yes	Yes

3 REQUIREMENTS ANALYSIS

3.1 Software Requirements

3.1.1 Python Libraries

Python Programming Language is the heart of the machine learning aspects of the project as well as parts of the backend. Python with its suite of various packages and libraries is used to facilitate the implementation of models for object detection and text recognition.

- **PyTorch and TensorFlow:** PyTorch and TensorFlow, both Deep Learning (DL) libraries, serve as the backbone for training and deploying machine learning models used in the project pipeline. They are used to first integrate existing models, specifically YOLOv7 and OCR, into the project and then also fine-tune and train them depending on the need. Also, the seamless integration of these libraries with other widely used Python libraries, including NumPy and Pandas, streamlines data preprocessing and manipulation. Both PyTorch and TensorFlow, while having different strengths, serve similar purposes for this project and use of one over the other just comes down to personal preference and convenience.
- **Numpy:** Numpy is an open-source Python library for all tasks related to numerical computing and data manipulation. CIRCUITRON uses Numpy for array manipulation and pixel-level operations on images. Numpy seamlessly integrates with OpenCV and helps streamline the fundamental processes of image processing like converting images into matrices, performing image-wide as well as element-wise operations important for augmentation, thresholding and segmentation.
- **OpenCV:** OpenCV is the primary library for pre-processing the circuit images and performing low-level computer vision tasks. It is used in almost all aspects of image processing in the project like thresholding, edge detection, contour detection, and color transformation as needed. These operations help in extracting meaningful visual information before feeding the data into the models.
- **Pandas:** Pandas provides data structures, such as data frames, that allow for efficient handling and manipulation of structured data. It offers functions for reading and writing data from various file formats, data cleaning and preprocessing, filtering and selecting data, grouping and aggregating data, and more. Pandas is used in the project to organize structured data related to the circuit components and their properties which includes storing component lists with attributes like label, value, position, and orientation, managing the intermediate data between detection, parsing, and netlist generation, and exporting tables or logs for debugging or evaluation.
- **Matplotlib and Plotly:** Matplotlib is a widely used data visualization library in Python. Matplotlib is used in the project for visualizing the performance metric scores

in graphs. Furthermore, it is to be used for plotting simulation outputs with integration with CircuitJS. Plotly is another library for data visualization which supports interactive diagrams (handy for showing simulation graphs in motion).

- **Scikit-learn:** In context of the project, Scikit-learn is used for post-processing tasks such as clustering junction points, filtering outliers, or classifying ambiguous component types after initial detection. It includes tools for data splitting, metrics, and evaluation which are extremely useful during model development and benchmarking. Additionally, it can assist in component matching, value prediction, or validation models if required.
- **Seaborn:** Seaborn is a Python data visualization library based on Matplotlib, designed for creating statistical graphics with attractive default styles. Circuitron uses this to visualize distributions, such as accuracy of component detection, or OCR confidence scores. It is essentially a wrapper for Matplotlib which makes working with graphs more convenient. It can also be used with Pandas to plot trends in simulation outcomes (say, voltage over time across components).
- **Hugging Face:** Hugging Face provides a large hub of pretrained transformer models for Natural Language Processing (NLP), vision, and more. It is used in the project to integrate LLMs for answering circuit-related queries and working with TrOCR.

3.1.2 YOLOv7

The YOLOv7 model is used to detect and classify circuit components in hand-drawn circuit images. It outputs bounding boxes, class labels, and confidence scores for each detected component. After comparison with other version of YOLO (v5, v8, v11), YOLOv7 is selected due to its superior detection accuracy, efficient training capabilities, and support for multiple object scales through its enhanced architecture, while also being of an appropriate size in regards to speed and training time.

3.1.3 OCR models

A CRNN-based OCR model trained from scratch, following the DTRB design philosophy for recognizing text is vital for the project as it is the core technology which reads and understands the component values and units of circuit elements. This has been chosen for the project because, foremost, other ready-made OCR models proved to be either too resource-hungry and/or failed to perform well even after some fine-tuning. Additionally, a minimally fine-tuned TrOCR model has also been used. The uses essentially gets to choose between a faster but slightly inferior custom OCR model, or a Microsoft-built robust yet slower TrOCR.

3.1.4 KiCad Symbol Library

KiCad is a free, open-source EDA suite which offers its libraries to be used freely to build any educational project. The project uses the KiCad symbols library, specifically, in order to simply extract the component symbols standardized by KiCad rather than building the icons/representations for components from scratch for use in the frontend schematic.

3.1.5 Coding and Training Platforms

- **Google Colab:** Google Colab is a cloud-based platform that offers a Jupyter notebook environment for developing deep learning systems. For this project, Google Colab provided with Graphics Processing Units (GPUs) for free, which significantly accelerated the training process, enabling experimentation with larger models and datasets. Colab also comes pre-installed with popular deep learning frameworks such as TensorFlow and PyTorch, widely used for Machine Learning (ML)-related tasks. Leveraging these frameworks, YOLO (v5 and v7) and OCR were implemented and experimented on.
- **Kaggle:** Kaggle is a multi-faceted online platform popular for data science competitions, datasets, code sharing and model training. It was primarily used to find relevant datasets (Circuit Graph Handwritten Diagrams (CGHD), VQA, Handwritten Circuit Diagram (HCD)), code sharing and model training. The dataset of choice for this project was also found in Kaggle. Additionally, it provided a fully hosted environment with popular Python libraries such as Pandas, Numpy, Tensorflow, PyTorch pre-installed to make implementing and training models convenient, especially with its integration with Google Colab. Its free cloud computing service providing 30 hours of GPU use per week proved indispensable in accelerating the training and testing of different models.
- **Lightning AI:** Lightning AI is a cloud workspace, similar to Google Colab and Kaggle, which specializes in helping build agents and train models without any tedious setup. Most importantly, through the use of a student account, it allows for the use of extremely powerful GPUs for free (for a reasonable amount of time). Among all the cloud computing services used for the project, Lightning AI proved the most invaluable for implementing and training the models as the GPUs it allotted were super-fast and saved a ton of time during the tedious and time-consuming process.
- **Visual Studio Code:** The local Integrated Development Environment (IDE) of choice for this project is VS Code due to its extensive features auto completion, syntax highlighting, and copilot integration. Additional features are further integrated by the use of a suite of easy-to-use extensions which are tailored for each individual project member in their setup as needed. All the code to be run locally is run on VS Code

and further down the line, its utility is expected to increase ten-fold during the web development phase.

3.1.6 Web Development

- **Reactjs:** Reactjs serves as an integral technology underlying the dynamic front-end interfaces to be developed for the project. The frontend for this project includes dynamic elements like dragging, erasing, resizing, etc. of the circuits and circuit elements, and Reactjs fulfills these requirements with convenience and speed through the use of its features like virtual Document Object Model (DOM) abstraction, distinction of code into reusable logic and presentation buckets, etc. as well as compatible packages like react-draggable.
- **Next.js:** Next.js is a React-based web framework that enables the development of fast, scalable, and server-rendered web applications. It provides features such as routing, server-side rendering, and Application Programming Interface (API) integration, which are essential for building a responsive and user-friendly frontend. In this project, Next.js is used to develop the client-side interface, enabling efficient interaction with the backend services and smooth visualization of recognition results.
- **FastAPI:** FastAPI is a modern, high-performance Python web framework used for building Representational State Transfer (REST) APIs. It is designed for rapid development, high execution speed, and automatic generation of interactive API documentation. In this project, FastAPI is used to implement the backend services that handle requests from the frontend, manage data exchange, and interface with the text recognition and analysis modules. Its native support for asynchronous operations and integration with Pydantic ensures efficient request handling and data validation.
- **Uvicorn:** Uvicorn is a high-performance Asynchronous Server Gateway Interface (ASGI) server used to run asynchronous Python web applications. It is commonly used to serve FastAPI-based backend services due to its low latency and efficient handling of concurrent requests. In this project, Uvicorn is responsible for hosting the backend API, ensuring fast communication between the frontend application and the OCR processing modules.
- **Pydantic:** Pydantic is a Python library used for data validation and schema definition using type annotations. It ensures that data exchanged between the frontend and backend conforms to predefined structures and types. In this project, Pydantic is used to define request and response models, improving data integrity, and reducing runtime errors.
- **Github:** Github is a platform to store, manage and share code with integrated version control. Github has enabled effective sharing and backing up of all the code involved

in this project and has allowed for all the participants to be on the same page at any given time.

3.2 Feasibility Study

3.2.1 Technical Feasibility

Technical Feasibility assesses whether the project can be developed using available resources and expertise. The system leverages well-established technologies such as Python, OpenCV, custom OCR, YOLOv7, FastAPI, Next.js, all of which are stable and widely supported. For component detection, YOLOv7 provides effective object recognition, making it suitable for identifying handwritten circuit symbols. Wire and node detection rely on proven techniques like skeletonization, junction-based mapping techniques, and clustering. Optical Character Recognition (OCR) is handled two-fold, effectively, on one hand using TrOCR and on the other using custom-trained OCR, ensuring accuracy even with noisy text. Simulation is facilitated through CircuitJS. The team consists of skilled electronics and software engineers proficient in these technologies. The primary technical risk involves processing low-quality or cluttered circuit images, but the system includes manual correction features to mitigate this. Overall, the project is technically feasible, with all necessary tools available and a well-matched skill set.

3.2.2 Economic Feasibility

Economic Feasibility examines whether the project is cost-effective and offers a good return on investment. The project costs include NPR 24,000 for computation (Google Colab Pro and cloud GPUs), NPR 2,160 for Institute of Electrical and Electronics Engineers (IEEE) membership (access to research journals), and NPR 15,000 for miscellaneous expenses (printing, presentations, memberships, etc.). Data collection and software licenses incur no cost since open-source datasets and tools are utilized. The total estimated budget is NPR 41,160. Given its potential as an educational tool, the project could even be commercialized for academic institutions. The project is considered economically feasible.

3.2.3 Operational Feasibility

Operational Feasibility evaluates whether the system will function effectively in real-world scenarios. The tool addresses a genuine need by simplifying the conversion of hand-drawn circuits into simulations, eliminating manual redrawing. Its user-friendly interface allows optional manual edits to improve reliability, and cross-platform accessibility is ensured through a web-based (Next.js and FastAPI) final product. The learning curve is minimal due to its intuitive design, and performance is optimized with

near real-time processing using YOLOv7 and efficient OCR techniques. The modular architecture ensures easy maintenance, and the system can be scaled to support additional features in the future. The project is considered operationally feasible.

3.2.4 Time Feasibility

The project has been developed in two sequential phases, with Phase A focusing on implementing the core functionality including the circuit recognition and text extraction engine, and netlist generation within a basic web framework. This phase was successfully completed and tested by September of this year.

Phase B has expanded the system by developing the full web application, integrating CircuitJS for simulations with a user-friendly UI to display and edit schematic diagrams, and implementing the AI question-answering feature; all while maintaining consistency with the pre-designed wireframes. A two-week buffer period has been utilized between phases to accommodate thorough testing and necessary adjustments, with Phase B work commencing only after Phase A components have been fully validated. This structured approach has ensured the maintenance of design integrity while progressively enabling the deliverance of more advanced features. As of yet, all of the promised features have been delivered with only minor kinks to be sorted out. The project is considered feasible from a time perspective.

4 SYSTEM ARCHITECTURE AND METHODOLOGY

4.1 Block Diagram

The process of converting a handwritten circuit diagram into a structured netlist involves a series of coordinated steps, each step feeding into the next, designed to detect, extract, and represent pictorial information in a machine-readable format and then using this data for simulation and schematic generation.

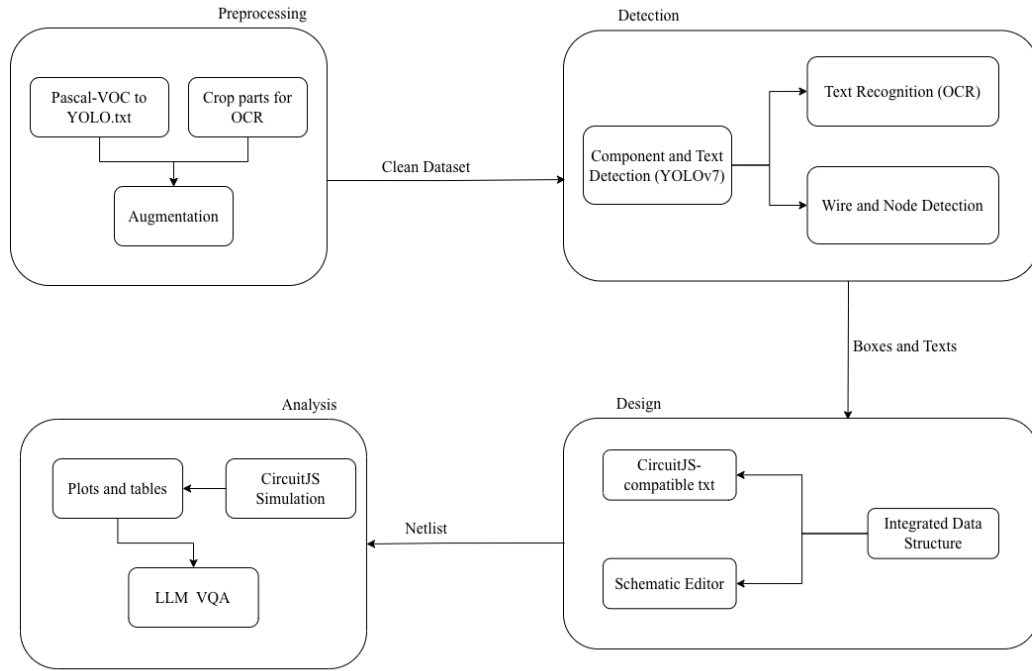


Figure 4-1 Block Diagram of the CIRCUITRON System

The high-level architecture of the CIRCUITRON system is visualized in the block diagram in Figure 4-1. Each step involved in the hand-drawn image to simulation-ready schematic pipeline is elaborated as follows.

4.1.1 Preprocessing

At first, the circuit information (components, component values, units, wire intersection, etc.) needs to be extracted out from the hand-drawn image. Appropriately fine-tuned ML and DL models are to be used for each of these steps which will be elaborated on in later subsections. However, for any sort of detection to happen, the image needs to first be pre-processed. Preprocessing for this not only includes image manipulation tasks but also conversion of the dataset annotations into appropriate formats to maintain compatibility with the component detection and text recognition models. The dataset of

choice is in the Pascal VOC format which is incompatible with the models used in the pipeline and hence need to be converted.

Furthermore, image preprocessing, which includes steps like thresholding, contrast and brightness enhancement (for improved clarity) is a vital step in readying the dataset for training any and all models. However, geometric augmentations, especially rotation, is not appropriate for this project.

Among the various available preprocessing techniques, the ones have been selected with careful consideration of the type of images available in the dataset as well as the models in use. Besides image manipulation, preprocessing also includes splitting the dataset into training, validation and testing subsets as appropriate.

4.1.2 Circuit Component Detection

A critical step in the pipeline is the accurate detection of circuit components from hand-drawn schematic images. This task is addressed using a real-time object detection algorithm capable of identifying and localizing multiple electronic components (like resistors, capacitors, voltage sources, etc.) directly within the image. YOLO algorithm, the YOLOv7 model, specifically, has been selected for the component detection needs of this project. It is to be noted that text, as a generic collection of letters, numbers, and special characters is also considered a component in this context.

YOLO (You Only Look Once) is a real-time object detection algorithm known for speed and efficiency. YOLO frames object detection as a single regression problem, straight from image pixels to bounding box coordinates and class probabilities. This model works as follows: first, dividing the image into an $S \times S$ grid. Each grid cell is responsible for detecting objects whose centre falls within it. For each cell, a number of bounding boxes and their corresponding confidence scores (which represents the accuracy of the box and whether the box contains an object) are predicted. Each predicted bounding box includes the coordinates of the box centre relative to grid cell, represented by (x, y) , the width and height relative to the whole image, represented by (w, h) and the confidence score C given as:

$$C = P(\text{object}) \cdot \text{IoU} \quad (4-1)$$

where $P(\text{object})$ denotes the probability of an object being inside the box, and Intersection over Union (IoU) is the Intersection over Union between the predicted box and the ground truth box.

Additionally, for each box, class probabilities are predicted as:

$$\text{Class Score} = P(\text{class } i \mid \text{object}) \cdot C \quad (4-2)$$

This allows the model to classify which object exists in each box and once the predictions are made, Non-Maximum Suppression (NMS) is applied to discard less-relevant overlapping boxes.

Based on similar fundamental working principle, several iterations (versions) of YOLO have been introduced and popularized among which YOLOv7 is the YOLO model of choice for this project.

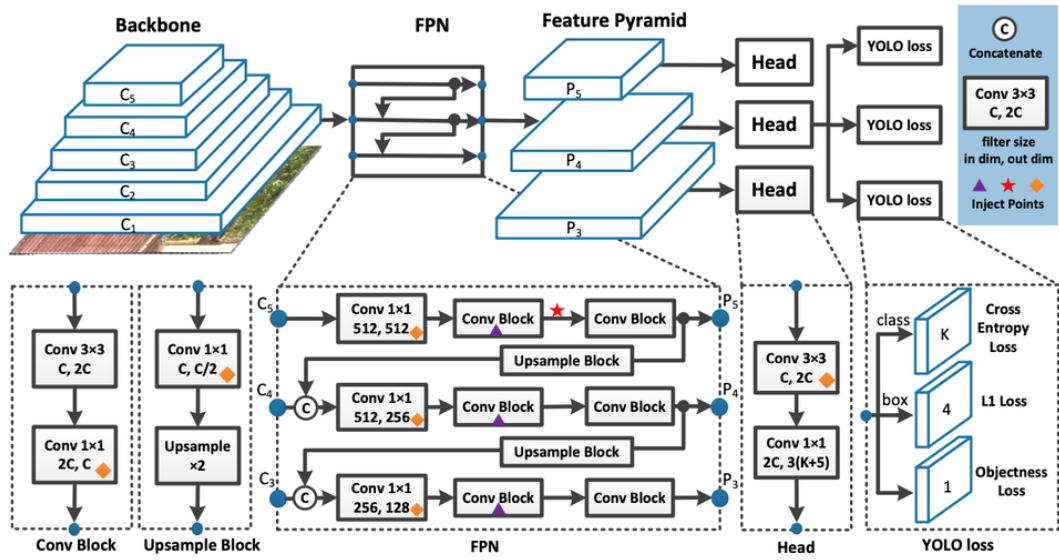


Figure 4-2 Architecture of YOLOv7

Figure 4-2 [37] shows the three primary modules of YOLOv7 architecture: Backbone, Neck, and Head.

- **Backbone:** YOLOv7 uses an extended version of ELAN as its backbone. This structure enables deeper network designs without significantly increasing computational cost. The backbone is responsible for extracting hierarchical features from the input image while preserving gradient flow and feature reuse.
- **Neck:** The neck in YOLOv7 incorporates a Path Aggregation Network (PAN) and ELAN-W modules. PAN enhances feature fusion across different scales, improving detection accuracy for both small and large objects. ELAN-W further refines spatial and contextual information during feature aggregation.
- **Head:** The head predicts object classes, bounding box coordinates (x, y, h, w), and objectness scores. YOLOv7 uses a decoupled head, which separates classification

and regression branches for better convergence and performance. The model outputs predictions at multiple scales to capture objects of varying sizes.

YOLOv7 introduces various upgrades over previous versions, while maintaining the size and speed advantages. The prediction of bounding box coordinates in YOLOv7 is guided by anchor-based regression, where the final bounding box $B = (x, y, w, h)$ is computed relative to the anchor box using the following formula:

$$x = \sigma(t_x) + c_x \quad (4-3)$$

$$y = \sigma(t_y) + c_y \quad (4-4)$$

$$w = p_w \cdot e^{t_w} \quad (4-5)$$

$$h = p_h \cdot e^{t_h} \quad (4-6)$$

where (t_x, t_y, t_w, t_h) are the predicted offsets, (c_x, c_y) is the top-left corner of the grid cell, and (p_w, p_h) are the dimensions of the anchor box.

YOLOv7 also introduces Coarse-to-Fine Lead Head, Model Scaling, and Auxiliary Heads that help improve gradient flow and stability during training. It supports Label Assignment Strategies such as dynamic-k matching for optimal anchor-label association. It applies automatic data augmentation during training including mosaic augmentation (combining four images into one), Hue, Saturation, Value (HSV) color augmentation, and random flipping, cropping, and scaling. These help increase dataset diversity and improve generalization. Likewise, it supports built-in early stopping based on a patience threshold if validation loss (combining box loss, objectness loss, and classification loss) does not improve for several epochs, ensuring faster convergence and no overfitting.

4.1.3 Recognition of Component Values

In order to generate a complete and simulation-ready netlist, it is essential to not only detect the components themselves but also to extract their associated parameter values, which are generally annotated near the components in the circuit diagram. Thus, it becomes essential to integrate Optical Character Recognition (OCR) with the object detection pipeline to extract and map these values correctly to the corresponding components.

This involves identifying regions within the circuit image that likely contain text (done by YOLOv7), and then recognizing said texts in order to parse the component values along with their units.

The hand-written text recognition is done using a CRNN-based OCR system, following the design philosophy of Deep Text Recognition Benchmark (DTRB) framework, however minor changes deemed suitable for this project have been made to get the text detection and recognition pipeline.

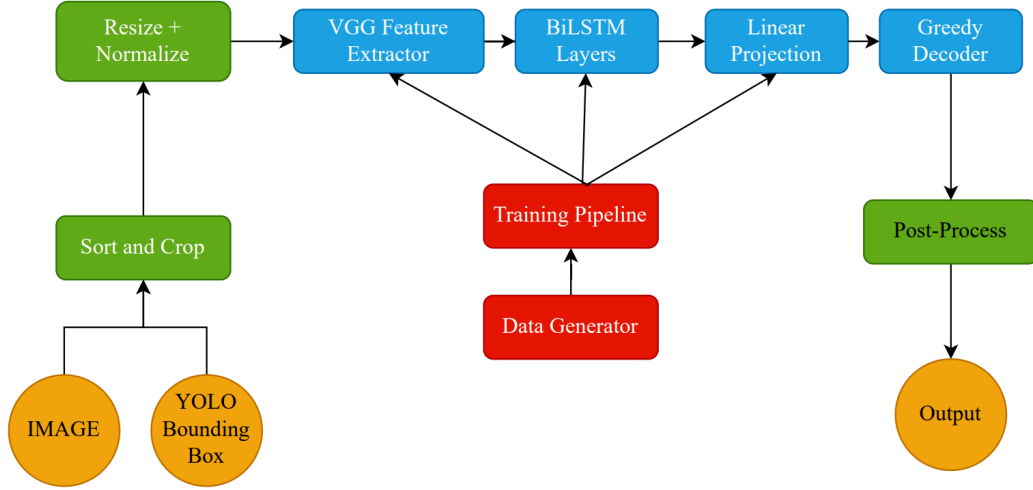


Figure 4-3 Text Detection and Recognition Pipeline

YOLOv7 outputs (bounding box information) help segregate textual information (class ‘0’ in the used dataset) which are then cropped, resized (to image height as 64) and normalized for compatibility with text recognition module. This is then passed through to a CRNN architecture composed of a Convolutional Encoder, Sequence Modeling Engine with Linear Projection, and CTC Decoder as shown in Figure 4-3.

Convolutional Encoder (CNN): A CNN stack (e.g., VGG) transforms input image I from multi-dimensional $\mathbb{R}^{H \times W \times C}$ to feature maps:

$$F = f_{\text{CNN}}(I_i), \quad F \in \mathbb{R}^{H' \times W' \times C} \quad (4-7)$$

Sequence Modeling (Bidirectional LSTM): Each F becomes input to a bidirectional LSTM, yielding hidden states h_t . These hidden states capture context across the text line, allowing modeling of character sequences:

$$\overrightarrow{h}_t = \text{LSTM}_f(F_t, \overrightarrow{h}_{t-1}) \quad (4-8)$$

CTC Decoder: A softmax layer projects h_t over the character vocabulary as:

$$p_{t,k} = \frac{e^{W_k h_t + b_k}}{\sum_{k'} e^{W_{k'} h_t + b_{k'}}} \quad (4-9)$$

The probability of sequence z is computed over all alignments:

$$P(z | y) = \sum_{\pi: B(\pi)=z} \prod_{t=1}^T p_{t,\pi_t} \quad (4-10)$$

The CTC loss is given by the negative logarithm of $P(z | y)$. It enables end-to-end sequence decoding without explicit character bounding boxes.

During inference, greedy decoding strategy is used which selects the most-probable character at each time step while removing blank symbols and collapsing consecutive duplicate characters. The recognized text then needs to be mapped to the appropriate circuit component whose value it represents.

This project includes two separate OCR models for character recognition, the second one being TrOCR (TrOCR-small to be exact) developed by Microsoft, specifically for handwritten text.

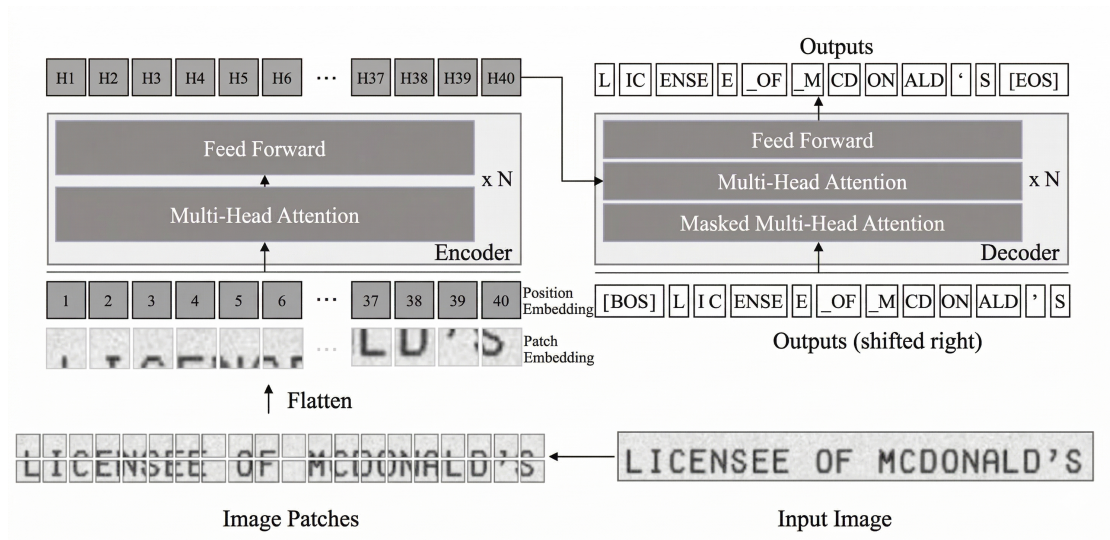


Figure 4-4 Architecture of TrOCR

Figure 4-4 [24] shows the encoder-decoder model (with pre-trained image transformer as encoder, and pre-trained text transformer as decoder) completing the basic architecture of TrOCR. The image transformer obtains the representation of image patches, and likewise,

with the help of visual features and previous predictions, the text transformer generates the wordpiece sequence.

The process begins with the raw input image, denoted as $x_{img} \in \mathbb{R}^{3 \times H_0 \times W_0}$. The transformer architecture is designed to process sequences rather than raw pixel grids, therefore the image is first resized to a standardized dimension (H, W) .

To format this data for the model, the image is decomposed into a series of N fixed-size square patches. Given a patch side length P , the total number of patches is derived as:

$$N = \frac{HW}{P^2}$$

Each patch is flattened into a vector and linearly projected into a D -dimensional space, where D represents the hidden dimension maintained throughout the transformer layers.

A [CLS] token is added to the start of the sequence to aggregate global representative information. When using pre-trained DeiT models for initialization, a distillation token is also included to facilitate learning from a teacher model. Since transformers lack the inherent spatial inductive bias found in CNNs, learnable 1D position embeddings are added to the patch embeddings to preserve absolute positional context. By treating the image as a sequence of independent patches, the model can dynamically allocate attention to either localized details or the global structure of the image.

The decoder utilizes the standard transformer decoder architecture to generate the final text sequence. While it shares a similar stack of identical layers with the encoder, it introduces an encoder-decoder attention block between the self-attention and the feed-forward networks. In this module, the queries originate from the decoder's internal state, while the keys and values are sourced from the encoder's output. This allows the decoder to selectively focus on relevant visual features during the generation process.

To prevent the model from accessing future tokens during training, a self-attention mask is applied. This ensures that the output for any position i is conditioned only on the preceding tokens. The decoder's hidden states are projected via a linear layer to the vocabulary size V . The final probability distribution over the vocabulary is calculated using the softmax function:

$$\sigma(h_{ij}) = \frac{e^{h_{ij}}}{\sum_{k=1}^V e^{h_{ik}}}$$

During inference, the model typically employs beam search to navigate the probability space and identify the most likely character sequence.

4.1.4 Wire Detection

Wire detection is performed using junction points detected by the object detection model as structural anchors for circuit connectivity. All detected junctions are first transformed from normalized detection space to pixel coordinates to ensure geometric consistency within the image domain. Pairwise combinations of junction points are then analyzed to identify potential wire candidates.

For each junction pair, a displacement vector and its orientation angle are computed. Based on a predefined angular tolerance, only near horizontal and near vertical orientations are retained, while diagonal connections are rejected to conform with standard schematic drawing conventions. Valid pairs are projected onto strict axis-aligned representations to form candidate wire segments.

To improve robustness against detection noise, wire endpoints are snapped to a uniform grid and collinear segments are consolidated using proximity and overlap criteria. This produces a compact and topologically consistent set of horizontal and vertical wire segments that represent the underlying circuit wiring.

4.1.5 Terminal Point Detection

Terminal point detection focuses on identifying meaningful electrical connection points associated with wires and components. Junction detections from the object detection model serve as primary terminal candidates, representing wire endpoints, intersections, and branching locations.

Once the final wire set is established, terminal points are refined by analyzing intersections between axis-aligned wire segments and component bounding boxes. Each intersection corresponds to a valid electrical terminal where a wire connects to a component edge. Intersection conditions are evaluated analytically to ensure precise localization of entry and exit points.

The resulting terminal points form a complete connectivity interface between wires and components, enabling accurate circuit graph construction and frontend mapping. This structured representation supports subsequent tasks such as netlist generation, interactive visualization, and automated circuit understanding.

4.1.6 Integration Algorithm

From the individual outputs from each of the aforementioned modules, the task is now to generate a structured JSON representation of the circuit which has is to be used to generate a corresponding CircuitJS-compatible text file and also contribute in the schematic generation further down the line.

The integration between all these different outputs, in a simplistic form, would consist of the following steps:

- Get a list of components and their corresponding bounding box coordinates (from YOLOv7).
- Get a list of text data and their corresponding bounding box coordinates (from OCR).
- Map the component to the corresponding component value (text) based on the minimum Euclidean distance between a particular component and texts on the circuit.
- Once the terminals are determined based on the bounding boxes, match those terminal points to the nearest nodes in the circuit.
- Store the accumulated information in a JSON file.

4.1.7 CircuitJS-compatible Text File Generation

A circuitJS-compatible text file describes the connections between components in an electronic circuit. It acts as a blueprint, specifying which components are connected and how, without detailing the physical layout or specific component placement.

It is a text-based representation of an electrical circuit used for simulation by the CircuitJS engine. It provides all the necessary information about the components, their connections, and their electrical properties, enabling analysis such as transient response, DC/AC sweep, or frequency domain behavior.

The text file serves as input to a circuit simulation engine, which then performs numerical computations to evaluate how the circuit behaves under specified conditions.

For each component, this text file contains the information about its name, the nodes it is connected to on either side and the component's parameters or model (if necessary).

After the JSON file with abundant information about the components, their values, and the nodes they are connected to is generated, the next step is to convert said JSON file into a CircuitJS-compatible file (a `.txt` file which follows the appropriate conventions).

4.1.8 Digital Schematic Generation

The JSON file built by integrating all the information about the hand-drawn circuit is to be manipulated through different JavaScript libraries to generate a digital, editable schematic of the circuit. As briefly discussed in the Requirements section, react-draggable is used to map each component in JSON to a visual block (a symbol for the component as given by the KiCad symbol library). For each component, a graphical symbol needs to be rendered at a certain position and connection lines need to be drawn between the terminals and node positions. A rough pseudocode for the same is as follows:

Algorithm 1 Frontend Circuit Rendering

Require: Circuit JSON description J

Ensure: Rendered circuit schematic

```
for all component  $c \in J.components$  do
    Draw schematic symbol of type  $c.type$  at position  $c.position$ 
    Attach text label  $c.value$  near component position
end for
for all connection  $k \in J.connections$  do
    Retrieve terminal position from  $k.from$  and terminal index
    Retrieve node position from  $k.node$ 
    Draw wire between terminal and node positions
end for
return Rendered schematic
```

4.1.9 Backend Development

The backend has been constructed using Next.js and FastAPI, functioning as the core middleware responsible for routing, and processing the system’s ML pipeline and user interactions. All endpoints follow a REST API design, ensuring stateless and scalable communication between the frontend and the server.

Upon receiving a hand-drawn circuit image through an API endpoint like `/upload`, the image is saved to a temporary storage location. The server invokes the YOLOv7 model (via a Python subprocess or containerized inference API) to detect and localize electrical components. Simultaneously, the OCR model is invoked to extract nearby textual annotations. The result is a list of components with bounding boxes and a list of recognized texts, each with confidence scores and coordinates.

Support for user interaction with the schematic is also handled by the backend. Operations such as component deletion, repositioning, wire addition, and modification of values are processed through specific API endpoints. Each update is stored in a temporary database, maintaining a real-time editable circuit representation.

4.1.10 LLM Integration

To enhance interaction with the circuit beyond GUI manipulation, a large language model (LLM), DeepSeek v3.1 to be exact is integrated. This makes the "analysis" part of the project a reality. It is responsible for handling user queries, such as, “Explain the purpose of this resistor.”, “Identify possible errors in this circuit.”, etc. The agent receives contextual data from the backend (such as the circuit’s JSON structure or CircuitJS file), formats the prompt, and forwards it to the LLM through a secure API.

4.2 Line Detection

Two approaches to line (wire) detection were developed and evaluated in this project. The first uses junction-point geometry with horizontal and vertical classification, while the second uses skeleton-based path finding with Multi-Head BFS. Both methods are compared in the results chapter; however, the BFS-based path-finding approach was adopted for the final frontend system due to its superior accuracy and robustness.

4.2.1 Line Detection via Horizontal and Vertical Classification

This approach detects wires by analyzing junction points identified by YOLOv7. The YOLO detection coordinates are first denormalized from the unit square $[0, 1]^2$ to pixel space using the affine map $\mathcal{N}^{-1}(x_n, y_n) = (W \cdot x_n, H \cdot y_n)$. The algorithm then performs pairwise analysis of all detected junction points: for each pair $(\mathbf{j}_i, \mathbf{j}_k)$, the displacement vector $\mathbf{d} = \mathbf{j}_k - \mathbf{j}_i$ is computed, and its angular orientation $\tilde{\theta}$ is obtained via the four-quadrant arctangent.

Junction pairs are classified using an 18° tolerance zone:

- **Horizontal:** $\tilde{\theta} \in [0^\circ, 18^\circ] \cup [162^\circ, 198^\circ] \cup [342^\circ, 360^\circ]$
- **Vertical:** $\tilde{\theta} \in [72^\circ, 108^\circ] \cup [252^\circ, 288^\circ]$
- **Diagonal (rejected):** all remaining angles (60% of angular space)

Valid axis-aligned pairs are projected onto perfectly horizontal or vertical lines by averaging the appropriate coordinate, then quantized to a 16-pixel grid. Overlapping collinear segments are consolidated via a merge algorithm that combines segments sharing the same orientation and lying within the grid tolerance.

Terminal detection under this approach identifies where wires connect to components by computing intersections between detected wire segments and component bounding boxes. For each wire-box pair, intersection conditions are evaluated at the relevant box edges (left/right for horizontal wires, top/bottom for vertical wires), yielding at most two intersection points per pair.

Shortcomings: This method has $O(n^2)$ pairwise complexity that scales poorly with junction count. The strict axis-alignment assumption rejects 60% of angular space, making it unable to handle angled or curved hand-drawn wires. Grid quantization introduces up to 11.3 pixels of positional error, and the consolidation heuristic can merge segments belonging to distinct wires in dense regions. Most critically, the method infers connections purely from junction positions with no knowledge of actual wire paths, producing spurious connections between unrelated axis-aligned junctions. The

terminal detection inherits these errors, as inaccurate wire segments directly produce false terminal points.

4.2.2 Line Detection via Path Finding

To address the limitations of the geometric approach, a skeleton-based path-finding method was developed. This pipeline operates directly on skeletonized circuit images, tracing actual wire paths rather than inferring connections from junction positions. It uses a Multi-Head Breadth-First Search (BFS) algorithm and consists of two stages: endpoint detection and graph construction.

Endpoint Detection: The strategy differs by object class. For junction points (class 1), a single endpoint is computed by snapping the bounding box center (x_c, y_c) to the nearest skeleton pixel within an expanded search region (margin = 8 pixels). For components (class 2+), two endpoints are computed by identifying skeleton pixels on or near the bounding box border, preferring true topological endpoints (degree = 1), and selecting the farthest pair.

Graph Construction via Multi-Head BFS: The algorithm constructs a graph where nodes are detected endpoints and edges represent skeleton connectivity. Each node occupies a disk of radius r , and a `node_id_map` is precomputed to associate each pixel with its owning node (or -1). For each source node, the BFS queue is initialized with skeleton pixels at the disk boundary and traverses along the skeleton using 8-connectivity. When a pixel belonging to another node j is encountered, the edge (i, j) is recorded and that branch terminates. After graph construction, self-edges (connecting endpoints of the same bounding box) are removed to ensure only inter-object connections remain.

Advantages over horizontal/vertical classification: The path-finding method traces actual wire paths in the image, eliminating spurious connections. It works regardless of wire orientation (horizontal, vertical, or diagonal). It unifies wire detection and terminal detection into a single graph-construction step, and scales linearly with the number of skeleton pixels rather than quadratically with junction count. Due to these advantages, this approach was adopted for the final deployed system.

4.2.3 Crossover Dissolution Algorithm

In hand-drawn circuit diagrams, a *crossover* represents two wires that cross each other without making an electrical connection, as opposed to a *junction* where wires meet and connect. Correctly distinguishing crossovers from junctions is critical for accurate circuit reconstruction. The YOLOv7 model detects crossovers as a dedicated class, and the Multi-Head BFS (Section 4.2.2) initially treats them as single-point graph nodes, typically connected to four neighboring endpoints corresponding to the two crossing

wire segments. The crossover dissolution algorithm resolves these nodes into their true topology.

Candidate Identification: Only crossover nodes with exactly four neighbors qualify for dissolution. Nodes with fewer neighbors are retained as regular junctions, gracefully handling cases where the detector misclassifies a junction as a crossover or where a wire terminates at the crossing point.

Direction Vector Computation: For each of the four neighbors $\mathbf{n}_k$ of a crossover node at position $\mathbf{p}$, a unit direction vector is computed as:

$$\hat{\mathbf{d}}_k = \frac{\mathbf{n}_k - \mathbf{p}}{\|\mathbf{n}_k - \mathbf{p}\|} \quad (4-11)$$

Optimal Pairing via Dot Product: The algorithm evaluates all three possible ways to partition the four neighbors into two pairs. For each candidate pairing $\{(\mathbf{n}_a, \mathbf{n}_b), (\mathbf{n}_c, \mathbf{n}_d)\}$, a score is computed as the sum of dot products within each pair:

$$S = \hat{\mathbf{d}}_a \cdot \hat{\mathbf{d}}_b + \hat{\mathbf{d}}_c \cdot \hat{\mathbf{d}}_d \quad (4-12)$$

The pairing with the minimum (most negative) score S is selected, as this maximizes the geometric opposition of paired neighbors, i.e., paired nodes lie on roughly opposite sides of the crossover.

Geometric Validation: The selected pairing is validated by checking that both dot products satisfy $\hat{\mathbf{d}}_a \cdot \hat{\mathbf{d}}_b < -0.7$, confirming that each pair of neighbors is within approximately 45° of being directly opposite through the crossover center.

Graph Transformation: If validation passes, the crossover node is removed from the adjacency graph and two new direct edges are created, one connecting each pair of opposite neighbors. This transforms a four-way junction into two independent through-connections, correctly representing two wires that cross without connecting. If validation fails, the node is preserved as a junction, maintaining electrical connectivity.

The dissolved graph is then used for final circuit assembly, where nodes represent component terminals, junctions, and wire endpoints, and edges represent the true wire paths, with crossover points absent from the electrical topology.

4.3 Use-Case Diagram

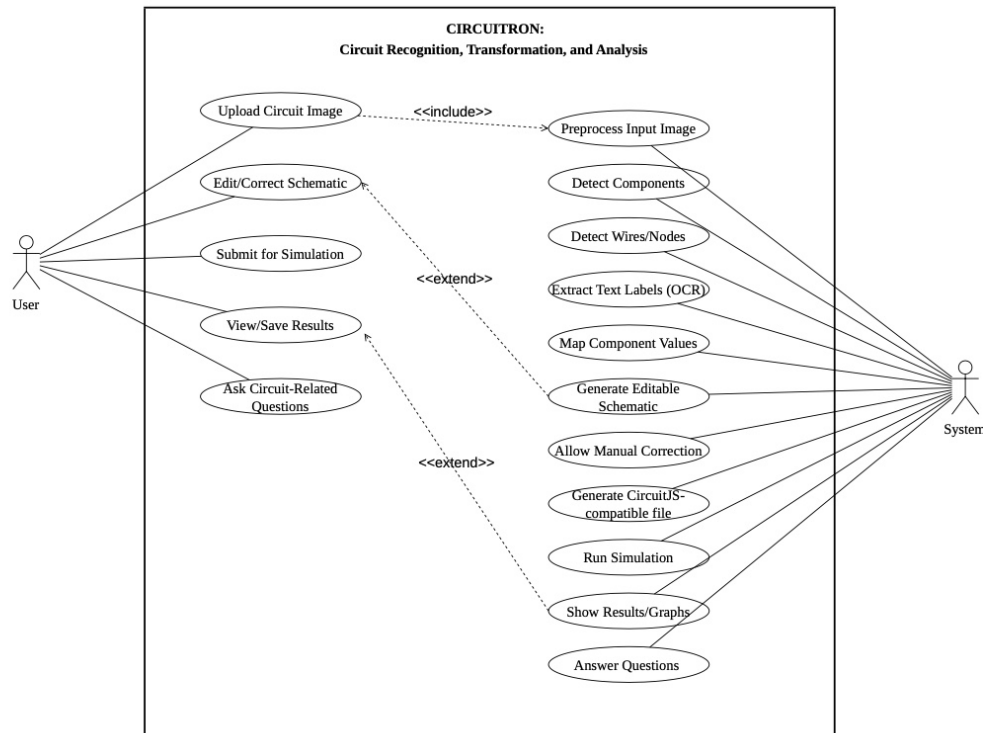


Figure 4-5 Use-Case Diagram for CIRCUITRON System

The use-case diagram in Figure 4-5 outlines the CIRCUITRON system’s functionality depicting the interactions between users and the system. The primary actor is the user, who can be a student, instructor, or engineer. The system begins with the user uploading a hand-drawn circuit image.

Upon submission, CIRCUITRON automatically preprocesses the image to enhance clarity and prepares it for further analysis. The system then detects components (e.g., resistors, capacitors) using YOLOv7 and identifies wires and nodes through techniques like junction-based wire detection. Simultaneously, it extracts text labels (e.g., component values) using OCR and maps them to the corresponding components.

Once the circuit structure is parsed, the system generates an editable schematic, allowing the user to manually correct any errors before proceeding. After validation, the system converts the schematic into a CircuitJS-compatible text file and runs a simulation using CircuitJS. The results, including voltage and current graphs, are displayed to the user. Additionally, the system integrates an AI assistant to answer circuit-related questions, enhancing the learning and troubleshooting experience.

The “includes” relationship is used for mandatory sub-tasks like component detection and OCR, which must always execute as part of core workflows. The “extends” relationship

marks optional features like manual corrections or AI Q&A, which enhance functionality but aren't required for basic operation. The diagram effectively outlines the system's scope, emphasizing its applications (or use-cases).

4.4 Performance Metrics Expectation

No matter how rigorously the models are trained and fine-tuned, there is always room for error due to edge cases like unclear handwriting and similar-looking components. As such, for any project, it is important to define appropriate performance metrics which indicate when the models can be considered “good-enough” for the required purpose. Table 4-1 shows the chosen performance metrics and their target values on achieving which, the training may be stopped.

Table 4-1 Expected Performance Metric Values for Individual Modules

Pipeline Stage	Performance Metric	Target Value/Expectation
Component Detection	mAP@0.5	0.95
Text Recognition	Character Accuracy	0.65

4.4.1 Component Detection Metric

mAP@0.5 (mean Average Precision at IoU threshold of 0.5) evaluates how well the model detects components with correct class labels and bounding boxes. It calculates the Average Precision (AP) for each class and then takes the mean across all classes. An IoU threshold of 0.5 means a predicted bounding box is considered a correct match if it overlaps with the ground truth box by at least 50%.

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (4-13)$$

A high mAP@0.5 means detections are both precise and complete, minimizing missed or misclassified components.

4.4.2 Text Recognition Metric

Character Accuracy measures how many characters in the predicted text match the actual characters:

$$\text{Character Accuracy} = \frac{\text{Number of Correct Characters}}{\text{Total Number of Characters in Ground Truth}} \quad (4-14)$$

In circuit diagrams, values like $10k$, 100Ω , $1\mu F$ are crucial. OCR must correctly read these values, since even a small error (e.g., mistaking ‘k’ as ‘1’) leads to incorrect

simulation behavior. A high character accuracy ensures value-to-component mapping is reliable.

4.5 System Flowchart

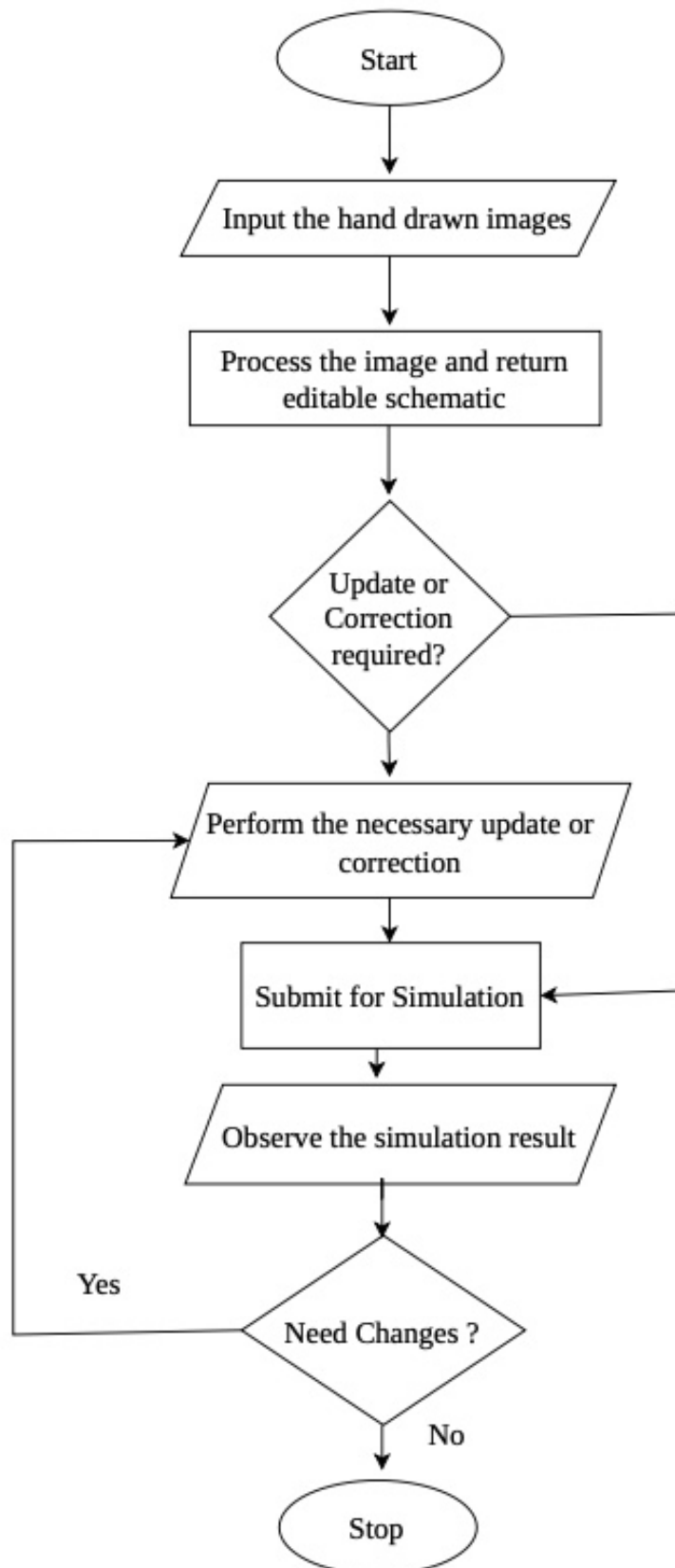


Figure 4-6 System Flowchart

The key architectural modules involved in the working of the system have been demonstrated by the block diagram in Figure 4-1. Now, the flowchart in Figure 4-6 demonstrates the motions a user needs to go through to use the final product. Once the image of a hand-drawn circuit is uploaded, the system processes the image (internally using the techniques explained in the previous sections) and returns an editable schematic. Any necessary update or correction may be performed if necessary and then the schematic can be submitted for simulation and analysis. Relevant graphs are returned after analysis.

4.6 Dataset Analysis

In order to develop the system using the system for automated circuit diagram understanding, the CGHD (Circuit Graph Handwritten Diagrams) dataset is used. This dataset provides a comprehensive suite of ground truth annotations tailored to facilitate key computer vision tasks such as object detection, instance segmentation, and text recognition. While additional datasets (such as circuitVQA, Digitize-HCD datasets) may be used to fine-tune the models in the future to achieve even higher values for performance metrics, as of yet, only the CGHD dataset has been used.

4.6.1 Dataset Overview

The CGHD dataset comprises 4.99 GB of annotated data spanning over 3,000 raw images. The data is separated into folders named drafters, going from -1 through 31, and each drafter folder contains hand-sketches of 12 unique circuit designs, with each circuit drawn twice and captured under four different conditions (varying angles, lighting, and minor distortions), resulting in up to 96 images per drafter. As is convention, the dataset is to be split into (70-10-20)% of the entire data for training, validation and testing respectively.

Table 4-2 Summary of Data in the CGHD Dataset

Items	Number
Distinct Circuits	372
Raw Images	3,269
Bounding Box Annotations	248,020
Text String Annotations	85,417 (93.74% completeness)
Character Types	98
Object Classes	61

Every image is rigorously annotated and the annotations provided per image include:

- **Bounding Boxes (PASCAL VOC format)** for 61 electrical component classes, including resistors, capacitors, transistors, operational amplifiers, and textual elements.

- **Text Labels** capturing component values and identifiers (e.g., “ $10k\Omega$ ”, “ $C1$ ”, “ V ”).
- **Instance Segmentation Polygons** (LabelMe JSON format) for pixel-level object outlines.
- **Binary Segmentation Maps** distinguishing foreground circuit strokes from background artifacts.
- **Partial Netlist Files** (in `.asc` format) for selected samples, useful for validating circuit reconstruction.

The text annotation coverage is extensive, with labels ranging from single characters (like “ R ” or “ $R1$ ”) to full descriptors (like “Resistor”, “Variable Resistor”, etc.).

4.6.2 Class-wise Data Statistics

Table 4-3 Frequency of Each Class with its Code in the CGHD Dataset

Code	Class Name	Count	Code	Class Name	Count
1	text	91,122	32	opamp.schmitt	16
2	junction	79,202	33	optocoupler	162
3	crossover	10,180	34	ic	1,506
4	terminal	11,253	35	ic.ne555	262
5	gnd	5,501	36	ic.v_regulator	362
6	vss	1,354	37	xor	162
7	voltage.dc	335	38	and	860
8	voltage.ac	292	39	or	440
9	voltage.battery	765	40	not	496
10	resistor	14,400	41	nand	228
11	resistor.adj	1,052	42	nor	59
12	resistor.photo	65	43	probe	184
13	cap.unpol	5,841	44	probe.curr	72
14	cap.pol	2,227	45	probe.volt	32
15	cap.adj	88	46	switch	2,184
16	inductor	1,976	47	relay	89
17	inductor.ferrite	120	48	socket	184
18	inductor.coupled	10	49	fuse	226
19	transformer	483	50	speaker	410
20	diode	3,890	51	motor	108
21	diode.led	1,212	52	lamp	301
22	diode.thyrector	16	53	microphone	72
23	diode.zener	609	54	antenna	24
24	diac	40	55	crystal	112
25	triac	98	56	mechanical	32
26	thyristor	415	57	magnetic	364
27	varistor	184	58	optical	127
28	transistor.bjt	3,678	59	block	0
29	transistor.fet	977	60	explanatory	610
30	transistor.photo	177	61	unknown	32
31	opamp	752			

5 IMPLEMENTATION DETAILS

The implementation of the system to convert hand-drawn circuit diagrams into simulation-ready netlists and digital schematics follows a modular approach, aligning with the pipeline described in the architecture and methodology.

5.1 Dataset Manipulation and Preprocessing

5.1.1 Conversion of Annotation Formats

To accommodate the multiple models in the pipeline (YOLOv7 for component detection and OCR for value extraction), annotation files provided in the CGHD dataset in Pascal VOC (XML) format were converted into compatible formats. Likewise, for EasyOCR-inspired custom-trained OCR, all text-sections needed to be cropped out from the image with corresponding labels (ground truth) and then fed to the fine-tuned model.

In Pascal VOC format, the information regarding the circuit and its components are stored in the following format as shown by the example below:

```
<annotation>
  <folder>images</folder>
  <filename>C1_D1_P2.jpg</filename>
  <path>./drafter_1/images/C1_D1_P2.jpg</path>
  <source>
    <database>CGHD</database>
  </source>
  <size>
    <width>3024</width>
    <height>4032</height>
    <depth>3</depth>
  </size>
  <segmented>0</segmented>
  <object>
    <name>voltage.dc</name>
    <pose>Unspecified</pose>
    <truncated>0</truncated>
    <difficult>0</difficult>
    <bndbox>
      <xmin>327</xmin>
      <ymin>546</ymin>
      <xmax>604</xmax>
      <ymax>838</ymax>
```

```

        <rotation>10</rotation>
    </bndbox>
</object>
</annotation>

```

For YOLO-based detection, XML annotations were converted to YOLO format (.txt) using a custom Python script. Each .txt file contains the class index and normalized bounding box coordinates in the format:

```
<class_id> <x_center> <y_center> <width> <height>
```

The pseudocode for the VOC to YOLO annotation conversion is as follows:

Algorithm 2 VOC to YOLO Annotation Conversion

Require: XML annotation file X , class map C , image width W , image height H

Ensure: List of YOLO formatted annotation lines Y

```

Parse XML file  $X$  into tree structure
 $Y \leftarrow \emptyset$ 
for all object nodes  $o$  in XML root do
    clsname  $\leftarrow$  class name of  $o$ 
    if clsname  $\notin C$  then
        continue
    end if
    clsid  $\leftarrow C[\text{clsname}]$ 
    Extract bounding box  $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$  from  $o$ 
    Convert VOC coordinates to normalized YOLO format
     $x_{\text{center}} \leftarrow \frac{x_{\min} + x_{\max}}{2W}$ 
     $y_{\text{center}} \leftarrow \frac{y_{\min} + y_{\max}}{2H}$ 
     $w \leftarrow \frac{x_{\max} - x_{\min}}{W}$ 
     $h \leftarrow \frac{y_{\max} - y_{\min}}{H}$ 
    Append formatted string  $(x_{\text{center}}, y_{\text{center}}, w, h)$  to  $Y$ 
end for
return  $Y$ 

```

The calculations run for this conversion is as follows:

$$x_{\text{center}} = \frac{327 + 604}{2 \times 3024} = \frac{465.5}{3024} = 0.1539 \quad (5-1)$$

$$y_{\text{center}} = \frac{546 + 838}{2 \times 4032} = \frac{692}{4032} = 0.1716 \quad (5-2)$$

$$w = \frac{604 - 327}{3024} = \frac{277}{3024} = 0.0916 \quad (5-3)$$

$$h = \frac{838 - 546}{4032} = \frac{292}{4032} = 0.0724 \quad (5-4)$$

This gives the information required by YOLO in the appropriate format. It is convention to save this new annotation file in the same directory as the image.

Likewise, for OCR compatibility, the same Pascal VOC annotations were converted to Lightning Memory-Mapped Database (LMDB) format. LMDB is a high-speed key-value store where each key corresponds to an index and each value contains an image-text pair (serialized).

The pseudocode for the conversion of Pascal VOC format to OCR-compatible LMDB file format is as follows:

Algorithm 3 Image Cropping and LMDB Dataset Creation

Require: Input image file I , crop coordinates $(y_{\min}, y_{\max}, x_{\min}, x_{\max})$, target size (W, H)

Ensure: LMDB database containing image-label pairs

```

Load image  $I$  into memory
Crop image using coordinates  $[y_{\min} : y_{\max}, x_{\min} : x_{\max}]$ 
Resize cropped image to size  $(W, H)$ 
Convert cropped image to byte representation
Convert image array to PIL image
Initialize in-memory byte buffer
Save image to buffer in JPEG format
Extract byte stream from buffer
Store image and label in LMDB
Open LMDB environment with predefined map size
Begin write transaction
Store image bytes with key image-00001
Store corresponding label with key label-00001
Store total sample count with key num-samples
Commit transaction
return LMDB environment

```

After this script is run in its entirety, a file in LMDB format, as is expected by OCR, is received as shown in the example below:

```

Key: b' image-00001'
Value: b'\xff\xd8\xff\xe0\x00\x10JFIF...'

Key: b' label-00001'
Value: b'voltage.dc'

Key: b' num-samples'
Value: b'000001'

```

After necessary conversion, the dataset was ready for being plugged into the required models for fine-tuning them. But first, some experimentation was done with different forms of image augmentation techniques.

5.1.2 Image Augmentation

Since the dataset contains handwritten circuit diagrams captured under variable lighting and orientations, several augmentation techniques were applied to improve model generalization including Contrast Enhancement (global) and Adaptive Brightness Adjustment by factors of 0.8 and 1.2. Furthermore, geometric augmentation techniques, like spatial flipping (both horizontally and vertically) and rotation (e.g., 90° or random rotation) proved to be unwise operations for this project. For example, if a “C” is rotated by 90°, the “C” now resembles a “U”, and the same applies to flipping (“M” might look like “W”, for example). Rather than increasing accuracy, it was realized that using rotation or flipping would simply confuse the model.

5.1.3 Fine-tuning YOLOv5 and YOLOv7 for Component Detection

YOLOv5 was initially used for benchmarking, and later upgraded to YOLOv7 for better accuracy and inference speed (comparison shown in the Results section). Training and fine-tuning were done using the CGHD dataset after converting into YOLO-compatible format.

The training parameters employed are as follows:

- **Number of classes:** 61
- **Training split:** 70% training, 10% validation, 20% testing
- **Epochs:** 200 (for YOLOv5), 100 (for YOLOv7)
- **Image size:** 416 × 416 pixels
- **Batch size:** 16 (for YOLOv5), 32 (for YOLOv7)
- **Optimizer:** Stochastic Gradient Descent (SGD) (Stochastic Gradient Descent)
- **Learning rate:** 0.01 (initial)
- **Augmentations used in training:** Mosaic Augmentation, HSV Augmentation, Random rotating/flipping

5.1.4 Class Consolidation and Retraining

Satisfactory results were not obtained through the initial 61-class approach. To improve the performance, consolidation of 61 classes into 15 was done, and the models were retrained on this new taxonomy.

Table 5-1 Consolidation of Original Classes into Unified YOLO Classes

Class Code	Class Name	Consolidated Classes
0	capacitor	capacitor.adjustable, capacitor.polarized, capacitor.unpolarized
1	crossover	crossover
2	diode	diode, diode.light_emitting, diode.thyrector, diode.zener
3	gnd	gnd
4	inductor	inductor, inductor.coupled, inductor.ferrite
5	integrated_circuit	integrated_circuit, integrated_circuit.ne555, integrated_circuit.voltage_regulator
6	junction	junction
7	operational_amplifier	operational_amplifier, operational_amplifier.schmitt_trigger
8	resistor	resistor, resistor.adjustable, resistor.photo
9	switch	switch
10	terminal	terminal
11	text	text
12	transistor	transistor.bjt, transistor.fet, transistor.photo
13	voltage	voltage.ac, voltage.battery, voltage.dc
14	vss	vss

A merge mapping was applied to the original annotations, grouping visually similar subclasses (e.g., capacitor.polarized and capacitor.unpolarized into capacitor) as shown in the Table 5-1. This consolidation is done not only considering visual similarity but also based on the usage i.e, the resulting 15 classes make up the majority of basic-level circuits. Four YOLO architectures were then trained on this 15-class dataset for benchmarking:

- **YOLOv7 (retrained):** 100 epochs, batch size 16, image size 640×640
- **YOLOv8:** 100 epochs, batch size 16, image size 640×640
- **YOLOv11:** 100 epochs, batch size 16, image size 640×640
- **YOLOv26:** 100 epochs, batch size 16, image size 640×640

All models were trained on identical data splits with the same augmentation pipeline, confidence threshold of 0.25, and IoU threshold of 0.45.

5.1.5 Implementation of Custom Text Recognition (OCR)

To extract textual information such as resistor values, capacitor labels, and other component annotations from the circuit diagrams, a CRNN-based custom OCR model was trained, and implemented. The input image in cropped form (cropped for each text component) was passed through this model after training, which returned the content of

each detected text element. The position of said text element was found using bounding boxes from YOLOv7. This allowed us to localize and read numerical values or labels directly from the circuit.

Three different modules, each for feature extraction, sequence modeling, and prediction were trained based on the DTRB framework. Initially, simply EasyOCR was also attempted; however, this didn't yield usable results, ultimately requiring training from scratch. The training, which was done for 27 epochs, had early stopping as it's accuracy per epoch failed to achieve even a 0.01% increase within 5 epochs. Other training details include:

- **Training epochs:** 27 (with early stopping)
- **Image Height:** 64px (fixed for all croppings)
- **Aspect Ratio:** 4:1 (allowing for variable width)
- **Learning Rate:** 0.01 (initial)

5.1.6 Fine-tuning TrOCR for Text Recognition

Following the initial custom CRNN approach, a second OCR model was implemented using Microsoft's TrOCR-small-printed architecture. Unlike the CRNN model which was trained from scratch, TrOCR was fine-tuned from pre-trained weights, leveraging transfer learning from large-scale printed text datasets.

The fine-tuning was performed on cleaned text crops extracted from the CGHD dataset. All text crops from a single image were processed in one forward pass, and per-token log-probability was averaged across the generated sequence with following details:

- **Base model:** Microsoft TrOCR-small-printed (pre-trained)
- **Fine-tuning epochs:** 3
- **Checkpoint:** Epoch 2 (best validation performance)
- **Inference precision:** float16 (half-precision for GPU acceleration)
- **Maximum output tokens:** 16 (sufficient for values like "10k Ω ", "100 μ F")

The TrOCR model is integrated into the pipeline through a singleton service that lazy-loads the processor and model weights when it is first invoked. This ensures efficient memory usage.

5.1.7 Line Detection Algorithm Implementation

The line detection mechanism was engineered to exploit the semantic constraints of circuit diagrams, specifically that wires are predominantly axis-aligned (horizontal or

vertical) and connect distinct components. Unlike the Probabilistic Hough Transform (PHT) which operates purely on pixel gradients, this approach uses the junction points detected by YOLOv7 as anchor nodes to reconstruct the wiring layout geometrically.

First, a pairwise analysis was performed on all junction points identified by the object detection model. The algorithm iterates through every unique pair of junctions to calculate the angle of the vector connecting them. Since hand-drawn circuits contain imperfections, an angular tolerance of $\pm 18^\circ$ was applied to classify connections as either horizontal or vertical. To ensure coordinate consistency across the diagram, the resulting segment coordinates were “snapped” to a grid using a 16-pixel quantization.

Algorithm 4 Wire Classification and Alignment

Require: Set of junctions J , angular tolerance $\omega_{\text{tol}} = 18^\circ$, grid size $g = 16$

Ensure: Set of raw wire segments S

```

 $S \leftarrow \emptyset$ 
for all pairs  $(j_a, j_b) \in \text{Combinations}(J, 2)$  do
     $\Delta x \leftarrow j_b.x - j_a.x$ 
     $\Delta y \leftarrow j_b.y - j_a.y$ 
     $\theta \leftarrow \text{atan2}(\Delta y, \Delta x)$ 
    if  $|\theta| \leq \omega_{\text{tol}} \vee |\theta - 180^\circ| \leq \omega_{\text{tol}}$  then
        Horizontal alignment check
         $y_{\text{avg}} \leftarrow \frac{j_a.y + j_b.y}{2}$ 
         $y_{\text{snap}} \leftarrow \text{Round}\left(\frac{y_{\text{avg}}}{g}\right) \times g$ 
         $S \leftarrow S \cup \{(H, y_{\text{snap}}, \min(j_a.x, j_b.x), \max(j_a.x, j_b.x))\}$ 
    else if  $|\theta - 90^\circ| \leq \omega_{\text{tol}} \vee |\theta - 270^\circ| \leq \omega_{\text{tol}}$  then
        Vertical alignment check
         $x_{\text{avg}} \leftarrow \frac{j_a.x + j_b.x}{2}$ 
         $x_{\text{snap}} \leftarrow \text{Round}\left(\frac{x_{\text{avg}}}{g}\right) \times g$ 
         $S \leftarrow S \cup \{(V, x_{\text{snap}}, \min(j_a.y, j_b.y), \max(j_a.y, j_b.y))\}$ 
    end if
end for
return  $S$ 

```

A common issue with the pairwise approach is the generation of redundant, overlapping segments for the same wire. For instance, if three junctions lie on a single line, the algorithm produces three separate segments (A-B, B-C, and A-C). To combat this, a collinear segment merging algorithm was implemented.

The segments were first separated by orientation (horizontal/vertical) and sorted by their primary axis. The merging logic then consolidated overlapping or adjacent segments. Crucially, a length-weighted heuristic was used: when merging, the axis coordinate of the longer segment was preserved, ensuring that larger, more reliable strokes dictated the final geometric alignment.

Algorithm 5 Collinear Segment Merging

Require: List of segments L , gap tolerance ω

Ensure: List of merged segments M

Sort L by primary axis, then by starting coordinate

$M \leftarrow \emptyset$

$\text{curr} \leftarrow L[0]$

for all $\text{next} \in L[1 \dots N]$ **do**

if $|\text{curr.axis} - \text{next.axis}| \leq \omega \wedge \text{next.start} \leq \text{curr.end} + \omega$ **then**

$\text{curr.start} \leftarrow \min(\text{curr.start}, \text{next.start})$

$\text{curr.end} \leftarrow \max(\text{curr.end}, \text{next.end})$

if $\text{Length}(\text{next}) > \text{Length}(\text{curr})$ **then**

Preserve axis of the longer segment

$\text{curr.axis} \leftarrow \text{next.axis}$

end if

else

$M \leftarrow M \cup \{\text{curr}\}$

$\text{curr} \leftarrow \text{next}$

end if

end for

$M \leftarrow M \cup \{\text{curr}\}$

return M

The wire–component intersection detection algorithm identifies precise connection points between detected wire segments and component bounding boxes. Component regions are first extracted from YOLOv7 annotations and converted into pixel coordinates. Each horizontal and vertical wire is then tested for geometric intersection with the edges of every component box. The resulting intersection points are used to accurately map circuit connectivity and are optionally visualized for verification in the frontend.

5.2 Line Detection via Path Finding

This section presents the algorithmic implementation of the line detection pipeline. The approach involves three key procedures: constructing a spatial mapping of node regions, performing multi-head BFS traversal to discover connectivity, and filtering out spurious self-edges.

The first step in the pipeline is building a node identification map. This data structure enables constant-time lookup to determine whether a given pixel belongs to a node region. For each detected endpoint, a circular disk of radius r is marked in the map, associating all pixels within that disk with the corresponding node index. Algorithm 7 formalizes this construction.

Algorithm 6 Wire–Component Intersection Detection

Require: Wire segments W , YOLO label file L , image width I_w , image height I_h

Ensure: Set of intersection points P

Parse label file L

Convert normalized bounding boxes to pixel coordinates

Discard non-component classes

Store component boxes B

Detect intersections between wires and components

$P \leftarrow \emptyset$

for all wire $w \in W$ **do**

for all box $b \in B$ **do**

if w is horizontal and crosses left or right edge of b **then**

 Add intersection point to P

else if w is vertical and crosses top or bottom edge of b **then**

 Add intersection point to P

end if

end for

end for

Draw wires, component boxes, and intersection points on image

Save output visualization

return P

Algorithm 7 Build Node ID Map

Require: Skeleton image S of size $H \times W$, list of nodes $\mathcal{N} = \{(x_0, y_0), \dots, (x_{n-1}, y_{n-1})\}$, radius r

Ensure: Node ID map M of size $H \times W$

1: $M \leftarrow$ array of size $H \times W$ initialized to -1

2: $\mathcal{D} \leftarrow \text{DiskMask}(r)$

3: **for** $i \leftarrow 0$ **to** $n - 1$ **do**

4: $(n_x, n_y) \leftarrow \mathcal{N}[i]$

5: **for** each $(dy, dx) \in \mathcal{D}$ **do**

6: $(p_y, p_x) \leftarrow (n_y + dy, n_x + dx)$

7: **if** $0 \leq p_y < H$ **and** $0 \leq p_x < W$ **then**

8: $M[p_y][p_x] \leftarrow i$

9: **end if**

10: **end for**

11: **end for**

12: **return** M

Once the node ID map is constructed, the algorithm performs a breadth-first search from each node to discover its neighbors along the skeleton. The BFS starts from the boundary of each node’s disk region and traverses skeleton pixels using 8-connectivity. When the traversal encounters a pixel belonging to a different node, that node is recorded as a neighbor and the branch terminates. This ensures that each skeleton path is followed until it reaches another endpoint. Algorithm 8 describes this procedure.

Algorithm 8 BFS Neighbors for Node

Require: Node index i , nodes $\mathcal{N}$, skeleton S , node ID map M , radius r

Ensure: Set of neighbor node indices $\mathcal{A}_i$

```
1:  $(n_x, n_y) \leftarrow \mathcal{N}[i]$ 
2:  $\mathcal{A}_i \leftarrow \emptyset$ 
3:  $V \leftarrow$  boolean array of size  $H \times W$  initialized to false
4:  $\mathcal{D} \leftarrow \text{DiskMask}(r)$ 
5: for each  $(dy, dx) \in \mathcal{D}$  do
6:    $(p_y, p_x) \leftarrow (n_y + dy, n_x + dx)$ 
7:   if  $0 \leq p_y < H$  and  $0 \leq p_x < W$  then
8:      $V[p_y][p_x] \leftarrow \text{true}$ 
9:   end if
10: end for
11:  $Q \leftarrow$  empty queue
12: for each  $(dy, dx) \in \mathcal{D}$  do
13:    $(p_y, p_x) \leftarrow (n_y + dy, n_x + dx)$ 
14:   for each  $(n'_y, n'_x) \in \text{Neighbors8}(p_y, p_x)$  do
15:     if  $\neg V[n'_y][n'_x]$  and  $S[n'_y][n'_x] > 0$  then
16:        $V[n'_y][n'_x] \leftarrow \text{true}$ 
17:        $Q.\text{enqueue}((n'_y, n'_x))$ 
18:     end if
19:   end for
20: end for
21: while  $Q$  is not empty do
22:    $(c_y, c_x) \leftarrow Q.\text{dequeue}()$ 
23:    $j \leftarrow M[c_y][c_x]$ 
24:   if  $j \neq -1$  and  $j \neq i$  then
25:      $\mathcal{A}_i \leftarrow \mathcal{A}_i \cup \{j\}$ 
26:     continue
27:   end if
28:   for each  $(n'_y, n'_x) \in \text{Neighbors8}(c_y, c_x)$  do
29:     if  $\neg V[n'_y][n'_x]$  and  $S[n'_y][n'_x] > 0$  then
30:        $V[n'_y][n'_x] \leftarrow \text{true}$ 
31:        $Q.\text{enqueue}((n'_y, n'_x))$ 
32:     end if
33:   end for
34: end while
35: return  $\mathcal{A}_i$ 
```

The final step removes spurious connections between endpoints that belong to the same detected object. Since line segment bounding boxes produce two endpoints each, the BFS may incorrectly connect these as neighbors via the skeleton segment within the bounding box. By maintaining a list of endpoint pairs from the same object, these self-edges are explicitly removed from the adjacency graph. Algorithm 9 implements this filtering.

Algorithm 9 Remove Self-Edges (Same-Object Connections)

Require: Adjacency $\mathcal{G}$, endpoint pairs $\mathcal{P} = \{(p_1^{(k)}, p_2^{(k)})\}_{k=1}^m$, nodes $\mathcal{N}$

Ensure: Filtered adjacency $\mathcal{G}'$

```
1:  $\mathcal{G}' \leftarrow \mathcal{G}$ 
2: for each  $(p_1, p_2) \in \mathcal{P}$  do
3:    $i \leftarrow \text{FindIndex}(p_1, \mathcal{N})$ 
4:    $j \leftarrow \text{FindIndex}(p_2, \mathcal{N})$ 
5:   if  $i \neq -1$  and  $j \neq -1$  then
6:      $\mathcal{G}'[i] \leftarrow \mathcal{G}'[i] \setminus \{j\}$ 
7:      $\mathcal{G}'[j] \leftarrow \mathcal{G}'[j] \setminus \{i\}$ 
8:   end if
9: end for
10: return  $\mathcal{G}'$ 
```

5.2.1 Complexity of Implemented Algorithm

This section analyzes the computational complexity of the line detection pipeline. Let H, W denote the image dimensions, n the number of nodes (endpoints), r the disk radius for node regions, and P the number of skeleton pixels (typically $P \ll H \times W$).

Table 5-2 Time complexity breakdown for Multi-Head BFS algorithm

Operation	Complexity	Description
Node ID Map Construction	$O(n \cdot r^2)$	Each node marks πr^2 pixels
Single Node BFS	$O(P)$	Each skeleton pixel visited at most once
Complete Adjacency Build	$O(n \cdot P)$	BFS from each of n nodes
Self-Edge Removal	$O(m)$	m = number of endpoint pairs
Total	$O(n \cdot r^2 + n \cdot P)$	Dominated by BFS traversals

Table 5-2 summarizes the time complexity of each operation in the pipeline.

Table 5-3 Space complexity breakdown for Multi-Head BFS algorithm

Data Structure	Space	Description
Skeleton Image S	$O(H \times W)$	Binary image storage
Node ID Map M	$O(H \times W)$	Integer array for node mapping
Visited Array V	$O(H \times W)$	Per-BFS boolean array
BFS Queue Q	$O(P)$	Maximum skeleton pixels in queue
Adjacency $\mathcal{G}$	$O(n + E)$	E = number of edges
Disk Offsets $\mathcal{D}$	$O(r^2)$	Precomputed disk mask
Total	$O(H \times W + n + E)$	Dominated by image arrays

Table 5-3 details the memory requirements of the algorithm.

The visited array can be reused across BFS calls with a reset, or alternatively, a hash set can be used for sparse skeletons to reduce visited storage to $O(P)$. For typical skeleton images where $P = O(H + W)$ (linear in image perimeter), the overall complexity is:

$$T(n, H, W, r) = O(n \cdot r^2 + n \cdot (H + W)) = O(n \cdot (r^2 + H + W))$$

In the real world, this approach can be considered efficient because it scales with the number of nodes, but it is bound by the physical resolution of the image sensor.

5.2.2 Crossover Dissolution

After the adjacency graph is constructed by Multi-Head BFS, crossover nodes must be resolved into their true topology. A crossover represents two wires crossing without connecting, and is initially represented as a single graph node with four neighbors. The dissolution algorithm identifies these nodes and replaces each with two direct through-connections, removing the node from the electrical topology.

Algorithm 10 Crossover Dissolution Algorithm

Require: Adjacency graph $\mathcal{G}$, node positions $\mathbf{p}$, detection results $\mathcal{R}$

Ensure: Modified graph $\mathcal{G}$ with crossovers dissolved

```
1: for each detection  $r \in \mathcal{R}$  with  $r.cls = 1$  do
2:    $a \leftarrow$  node index of  $r$ 's endpoint in  $\mathbf{p}$ 
3:   if  $a = -1$  or  $|\mathcal{G}[a]| \neq 4$  then
4:     continue
5:   end if
6:    $\{n_1, n_2, n_3, n_4\} \leftarrow \mathcal{G}[a]$ 
7:   for each neighbor  $n_k$  do
8:      $\hat{\mathbf{d}}_k \leftarrow \frac{\mathbf{p}[n_k] - \mathbf{p}[a]}{\|\mathbf{p}[n_k] - \mathbf{p}[a]\|}$ 
9:   end for
10:   $S^* \leftarrow +\infty$ ,  $\Pi^* \leftarrow \emptyset$ 
11:  for each pairing  $\Pi = \{(n_i, n_j), (n_k, n_l)\}$  of the 4 neighbors into 2 pairs do
12:     $S \leftarrow \hat{\mathbf{d}}_i \cdot \hat{\mathbf{d}}_j + \hat{\mathbf{d}}_k \cdot \hat{\mathbf{d}}_l$ 
13:    if  $S < S^*$  then
14:       $S^* \leftarrow S$ ,  $\Pi^* \leftarrow \Pi$ 
15:    end if
16:  end for
17:   $\{(n_a, n_b), (n_c, n_d)\} \leftarrow \Pi^*$ 
18:  if  $\hat{\mathbf{d}}_a \cdot \hat{\mathbf{d}}_b > -0.7$  or  $\hat{\mathbf{d}}_c \cdot \hat{\mathbf{d}}_d > -0.7$  then
19:    continue
20:  end if
21:  for each neighbor  $n_k \in \{n_a, n_b, n_c, n_d\}$  do
22:     $\mathcal{G}[n_k] \leftarrow \mathcal{G}[n_k] \setminus \{a\}$ 
23:  end for
24:  Remove  $a$  from  $\mathcal{G}$ 
25:  Add edges  $(n_a, n_b)$  and  $(n_c, n_d)$  to  $\mathcal{G}$ 
26: end for
27: return  $\mathcal{G}$ 
```

The algorithm evaluates all three possible pairings of the four neighbors and selects the one that minimizes the sum of dot products, ensuring paired neighbors are maximally opposite. The threshold of -0.7 (corresponding to approximately 45° from direct opposition) validates that the pairing represents a genuine crossing geometry. Crossover nodes that fail validation (fewer than 4 neighbors or non-opposing geometry) are retained as regular junctions.

5.3 Frontend and Backend Implementation

The frontend system has been realized using React.js, CircuitJS and relevant packages like react-draggable and KiCad Symbol Library. Likewise, the backend of the circuitron system exposes a REST API that manages the complete lifecycle of circuit simulations. The API provides endpoints for submitting CircuitJS-compatible text files, validating

circuit syntax, and executing simulations using different analysis modes such as transient, AC, and DC analysis.

Each simulation request is assigned a unique identifier, enabling asynchronous execution and progress tracking through dedicated status endpoints. The API also supports retrieval of detailed simulation results, including time-domain voltage and current data, as well as administrative operations such as stopping running simulations, listing historical simulations, and deleting completed or failed runs. Standard Hypertext Transfer Protocol (HTTP) status codes and structured JSON responses are used throughout the API to ensure consistency, robust error handling, and seamless integration with the frontend application.

5.3.1 Frontend-Backend Interaction

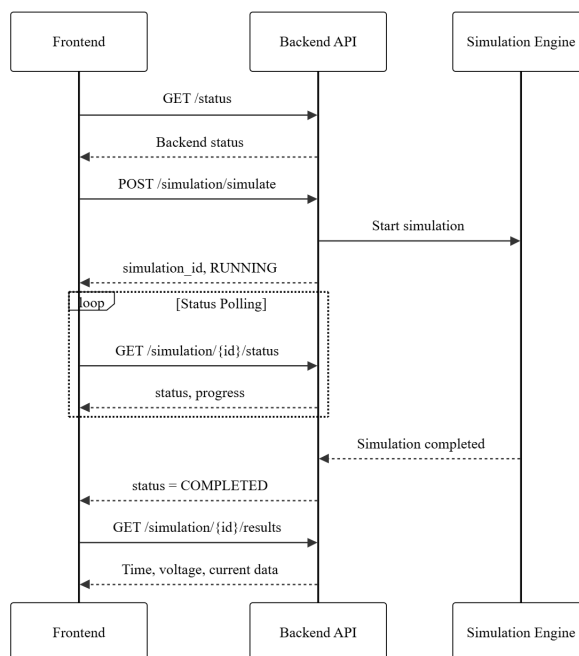


Figure 5-1 Interaction Diagram for the Web Application

The frontend communicates with the backend through HTTP requests to the documented endpoints. User inputs from the editable circuit schematic are converted into CircuitJS-compatible text files and submitted to the simulation endpoint. The frontend periodically polls the status endpoint to track simulation progress, and upon completion, retrieves and visualizes the results using CircuitJS for graph generation and React components for interactive display.

5.4 Circuit Simulation using CircuitJS

CircuitJS converts the scanned circuit into matrix-based mathematical equations. This means instead of drawing wires, internally, it builds a numerical model (network graph) of the circuit for simulation. These matrix-based mathematical equations are solved repeatedly in a loop, which is called time-domain simulation. For non-linear components, iterative methods are used until convergence. The integration of CircuitJS into the frontend enables seamless interaction with the simulation backend, allowing users to design circuits, run simulations (visualized with animation), and analyze results without requiring any additional software installations.

5.5 Error Handling

In any project with a number of individual modules working together, the overall error not only comes from said individual modules but also from their integration due to propagating errors from one module to another. To handle errors in such a case, foremost the errors in the output of the individual modules need to be controlled and only then can the accumulated error be kept in check. The performance metrics tabulated in the previous section are to be strictly adhered to if there's any hope of generating the CircuitJS compatible text file (and digital schematic) with nominal errors.

For the reduction of false matches and OCR noise, post-processing techniques such as value format validation, confidence filtering, and text cleanup can be implemented. Value format validation ensures only strings that match known (and appropriate) electrical units, say $10k$, $1\mu F$, $5V$ are retained. Confidence filtering helps discard OCR results with low confidence scores, and text cleanup normalizes ambiguous symbols such as converting “O” to “0”, “ μ ” to “u”.

A basic pseudocode for sample value format validation is as follows:

Algorithm 11 Validation and Filtering of OCR Component Values

Require: OCR detected text entries T

Ensure: Filtered list of valid component values V

Define regular expression patterns for resistance, capacitance, and voltage

$V \leftarrow \emptyset$

for all text entry $t \in T$ **do**

if t .text matches any defined pattern **then**

 Append t to V

end if

end for

return V

The list of components shown here is in no way meant to be exhaustive; rather it is to show how this pseudocode ensures that only valid electrical component values are used in the final component-to-value mapping and schematic generation.

To prevent propagating error (error from the output of YOLOv7 bleeding into OCR, for instance), a combined confidence threshold checking mechanism can be implemented which marks an item as uncertain if the confidence score from YOLOv7 is less than a certain value (e.g., 0.6) and the same from OCR is less than a certain value (e.g., 0.85). For more robustness, a human-in-the-loop mechanism can be implemented which prompts the user to manually approve or correct OCR-predicted values, fix connection errors, etc. This can be implemented to trigger when the confidence score is below the set threshold or when other error handling mechanisms such as value format validation fail. The pseudocode for this is as follows:

Algorithm 12 Confidence-Based Error Checking and Validation

Require: Module outputs O , confidence threshold ω , format validation function f

Ensure: Valid outputs V and uncertain outputs U

```

 $V \leftarrow \emptyset$ 
 $U \leftarrow \emptyset$ 
for all item  $o \in O$  do
    if  $o.\text{confidence} \geq \omega \wedge f(o.\text{value})$  then
        Append  $o$  to  $V$ 
    else
        Append  $o$  to  $U$ 
    end if
end for
if  $U \neq \emptyset$  then
    Trigger manual override
end if
return  $(V, U)$ 

```

6 RESULTS AND ANALYSIS

6.1 Comparative Analysis of Object Detection using YOLOv5 and YOLOv7

6.1.1 Experimental Setup

A comparative study using YOLOv5 and YOLOv7 to detect and classify hand-drawn electronic circuit components was conducted to determine which model is better-suited for the needs of this project. The dataset consisted of 61 classes, including resistors, capacitors (polarized and unpolarized), diodes, transistors (Bipolar Junction Transistor (BJT) and Field-Effect Transistor (FET)), integrated circuits, switches, and more. To ensure a fair comparison, both models were trained and validated on identical data splits.

6.1.2 Performance Metrics

mAP vs Number of Epochs

In order to monitor the progress in models' ability to recognize components as the training progressed, mAP@0.5 was plotted across all epochs. These curves depict how quickly and effectively each model learns over time.

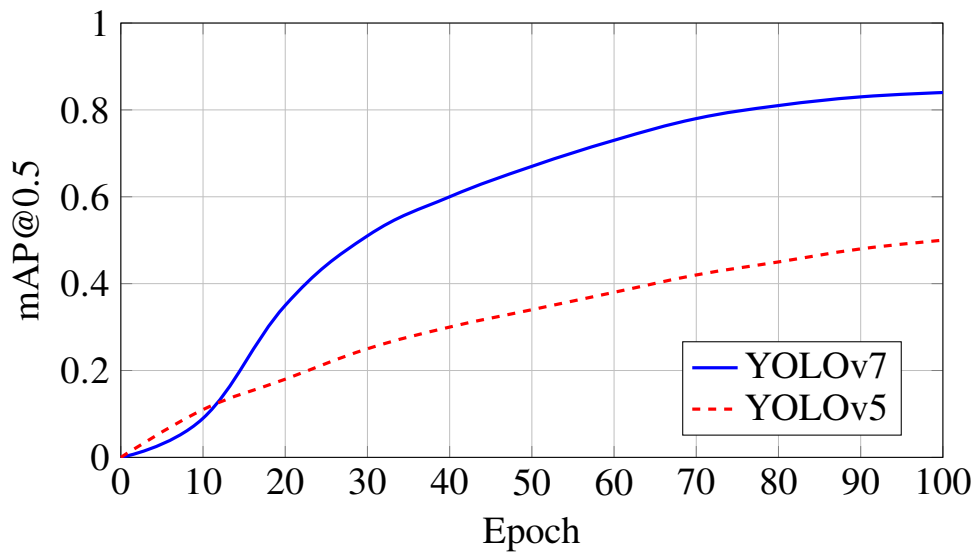


Figure 6-1 mAP@0.5 comparison over 100 training epochs.

It is clear from Figure 6-1 that the mAP score for YOLOv7 are significantly better than that for YOLOv5.

Precision vs Recall and F1 Score vs Confidence

Figure 6-2 demonstrate the F1 Score-Confidence Curve for YOLOv5 and YOLOv7 respectively. The larger the area under this curve, the higher the robustness of the precision-recall trade-off across thresholds.

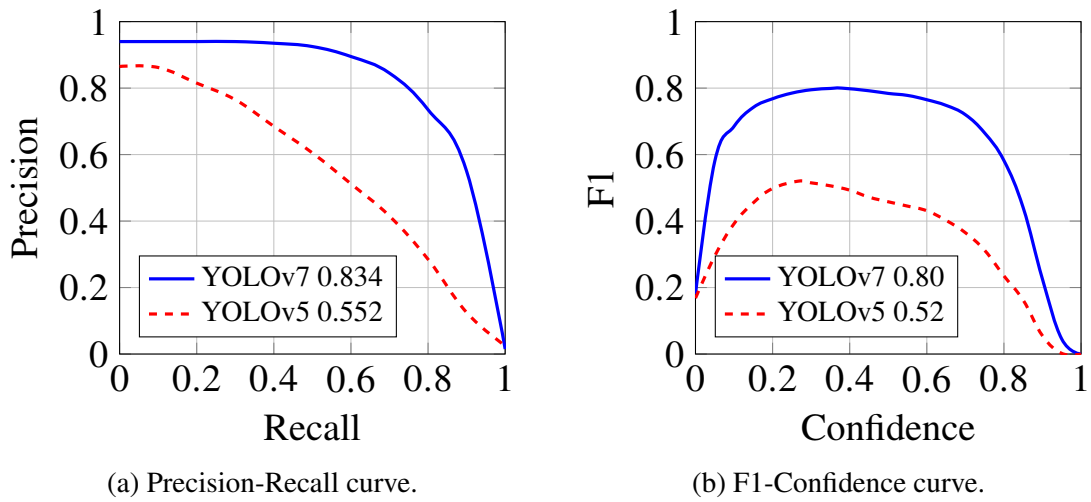


Figure 6-2 Performance metrics comparison: (a) Precision-Recall and (b) F1-Confidence curves.

Figure 6-2 also demonstrate the Precision-Recall Curve for YOLOv5 and YOLOv7 respectively. The area under this curve gives the average precision (AP) which informs how good the model is at ranking positive samples above negative ones. The area under each of the curves in these figures is higher for YOLOv7 in comparison to YOLOv5. This is expected behavior and proves YOLOv7 is better than YOLOv5 when it comes to performance.

6.1.3 Quantitative Results Comparison

Final epoch results show YOLOv7 outperformed YOLOv5 across all major metrics.

Table 6-1 Comparison of Performance Metrics for YOLOv5 and YOLOv7

Metric	YOLOv5	YOLOv7
mAP@0.5	0.61	0.83
mAP@0.5:0.95	0.41	0.60
Precision	0.84	0.93
Recall	0.62	0.78
Epochs Trained	200	100

Overall, YOLOv7 showed a clear advantage over YOLOv5 in our experiments. It not only achieved higher values for precision, recall, and mean average precision but also required fewer epochs to converge. YOLOv7's confusion matrix reflected better class separation and fewer misclassifications. The improvements in F1 score and overall stability of predictions across confidence thresholds make YOLOv7 a more robust choice for circuit diagram interpretation.

6.2 Class Taxonomy Consolidation

A key finding from the initial 61-class experiments was that many fine-grained component subclasses (e.g., capacitor.polarized, capacitor.unpolarized) shared near-identical visual features, causing inter-class confusion. To address this, the 61-class taxonomy was consolidated into a 15-class merged taxonomy by grouping visually similar subclasses. This section presents findings from training and evaluating four YOLO architectures (YOLOv7, YOLOv8, YOLOv11, YOLOv26) on the consolidated 15-class dataset along with comparative analysis with the 61-class results.

It is to be noted that, for fairness, all four YOLO architectures were trained and evaluated on the 15-class dataset under uniform conditions. All models used identical training/validation/test splits, a confidence threshold of 0.25, and an IoU threshold of 0.45 during evaluation.

6.2.1 Training and Validation Loss: 15-Class Models

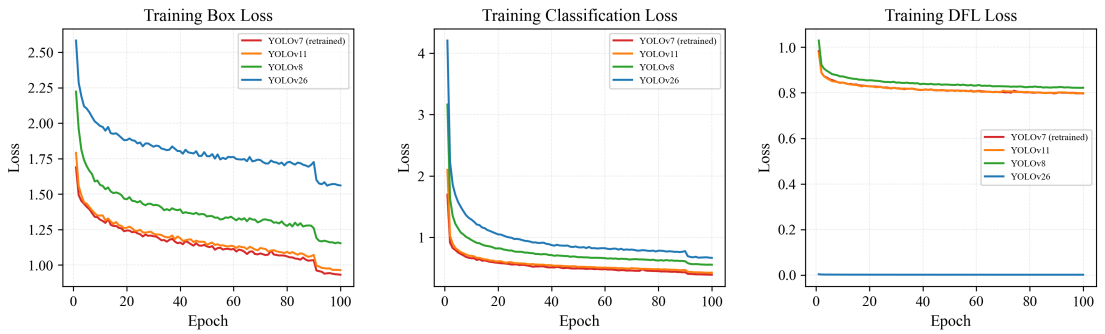


Figure 6-3 Training loss comparison across four YOLO architectures

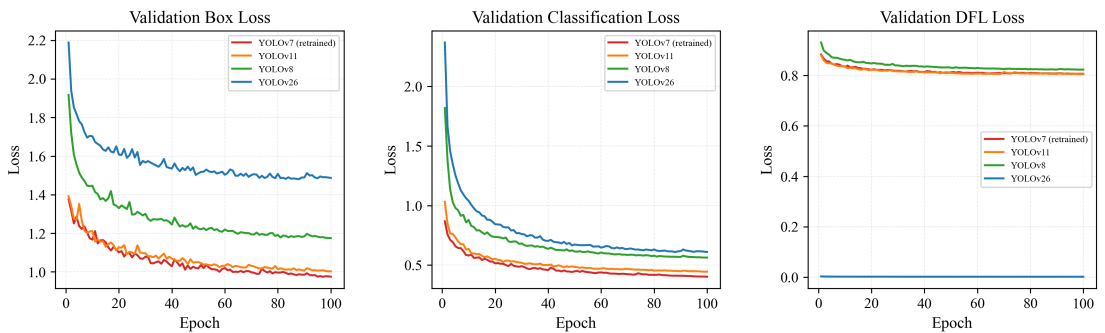


Figure 6-4 Validation loss comparison across four YOLO architectures

Figure 6-3 and Figure 6-4 present the training and validation losses (box, classification, and DFL) for all four models. YOLOv7 (retrained) and YOLOv11 reach lower training losses more rapidly, indicating superior representational capacity. Validation losses follow a similar trend, with YOLOv7 and YOLOv11 achieving the lowest final validation

losses, confirming that these models generalize better and are less prone to overfitting on this dataset.

6.2.2 Performance Metrics: 15-Class Models

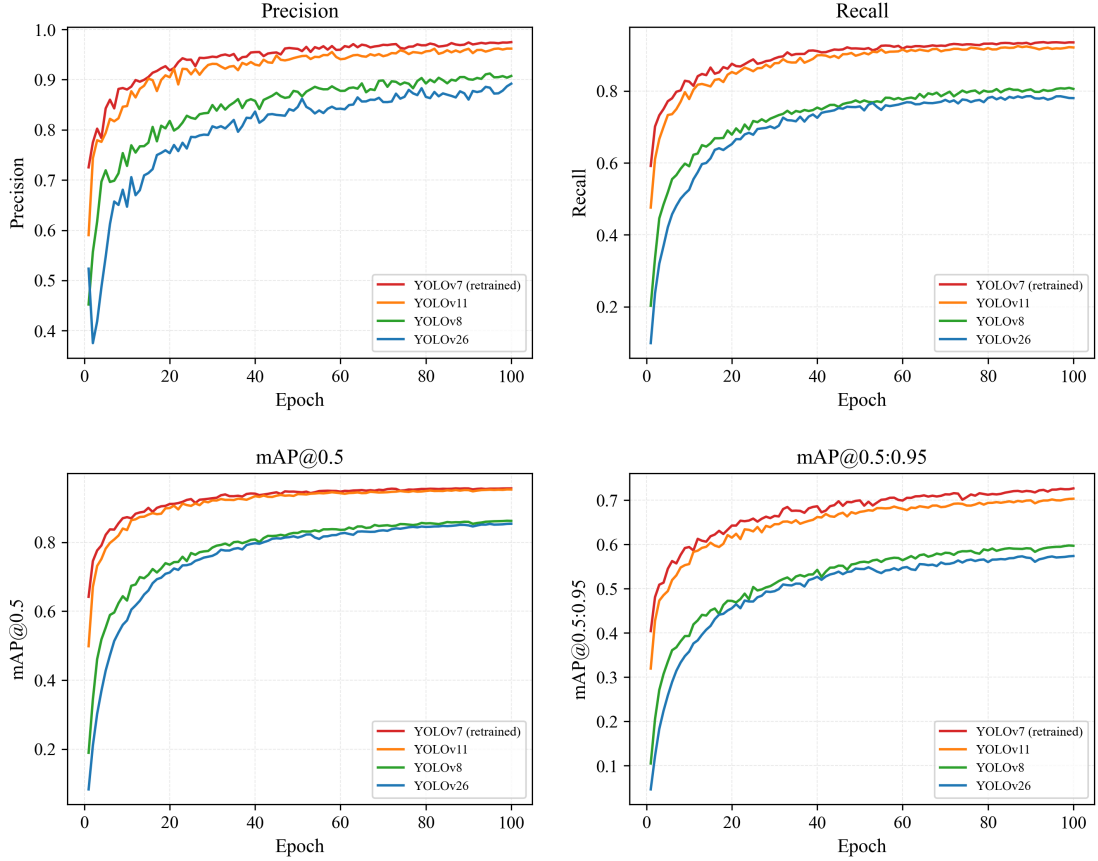


Figure 6-5 Performance metrics vs. epochs for four YOLO architectures on the 15-class dataset

Figure 6-5 shows the evolution of all four performance metrics across 100 training epochs for the four architectures. Both YOLOv7 (retrained) and YOLOv11 converge rapidly, reaching near-peak performance by epoch 40–50 and maintaining stable values thereafter. In contrast, YOLOv8 and YOLOv26 exhibit slower convergence with substantially lower plateaus, particularly for recall and mAP@0.5:0.95.

6.2.3 Quantitative Results Comparison for 15-Class Dataset

Table 6-2 Performance comparison of four YOLO architectures on the 15-class dataset (final epoch)

Metric	YOLOv7 (retrained)	YOLOv11	YOLOv8	YOLOv26
Precision	0.9747	0.9617	0.9073	0.8919
Recall	0.9350	0.9209	0.8058	0.7801
mAP@0.5	0.9562	0.9531	0.8620	0.8537
mAP@0.5:0.95	0.7266	0.7033	0.5969	0.5738
Epochs	100	100	100	100

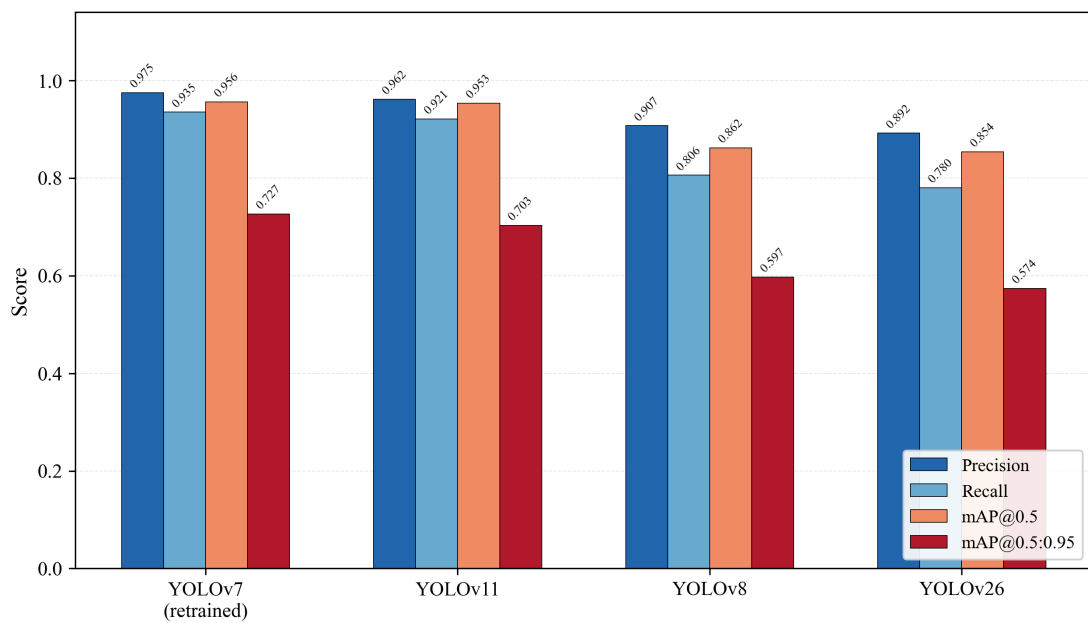


Figure 6-6 Final-epoch performance metrics for four YOLO architectures on the 15-class dataset

Table 6-2 summarizes the final-epoch performance of all four architectures on the 15-class dataset, and Figure 6-6 provides a grouped bar chart visualization. YOLOv7 (retrained) achieves the highest scores across all four metrics, closely followed by YOLOv11. Both models surpass 95% mAP@0.5, demonstrating exceptional detection capability on the consolidated class set. YOLOv8 and YOLOv26, while adequate, lag behind by approximately 10 percentage points in mAP@0.5 and 13 percentage points in mAP@0.5:0.95, making them less suitable for this project.

6.2.4 Impact of Class Taxonomy Consolidation

Table 6-3 Impact of class taxonomy consolidation on YOLOv7 performance

Metric	61-Class	15-Class	Improvement
Precision	0.93	0.9747	+4.8%
Recall	0.78	0.9350	+19.9%
mAP@0.5	0.83	0.9562	+15.2%
mAP@0.5:0.95	0.60	0.7266	+21.1%

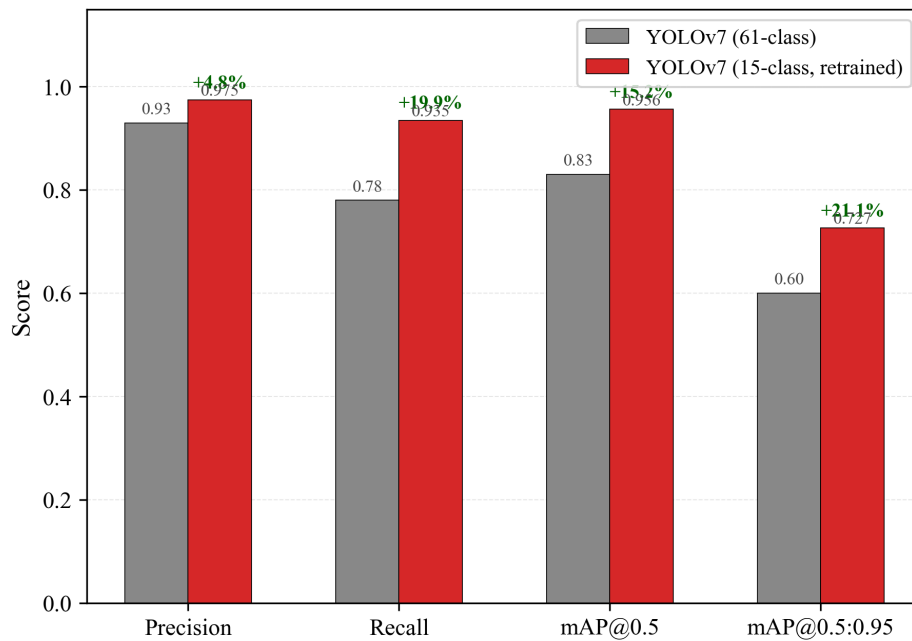


Figure 6-7 Effect of class taxonomy consolidation on YOLOv7: 61-class vs. 15-class performance

To quantify the effect of reducing 61 classes to 15, the same architecture (YOLOv7) is compared under both taxonomies. Figure 6-7 and Table 6-3 show the results. Consolidation yielded substantial gains: precision improved by 4.8%, recall by 19.9%, mAP@0.5 by 15.2%, and mAP@0.5:0.95 by 21.1%. These improvements confirm that the original fine-grained taxonomy introduced unnecessary inter-class confusion that degraded detection performance.

6.3 Holistic Object Detection Comparison

Table 6-4 Unified comparison of all object detection configurations

Model	Classes	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv5	61	0.84	0.62	0.61	0.41
YOLOv7	61	0.93	0.78	0.83	0.60
YOLOv8	15	0.9073	0.8058	0.8620	0.5969
YOLOv26	15	0.8919	0.7801	0.8537	0.5738
YOLOv11	15	0.9617	0.9209	0.9531	0.7033
YOLOv7 (retrained)	15	0.9747	0.9350	0.9562	0.7266

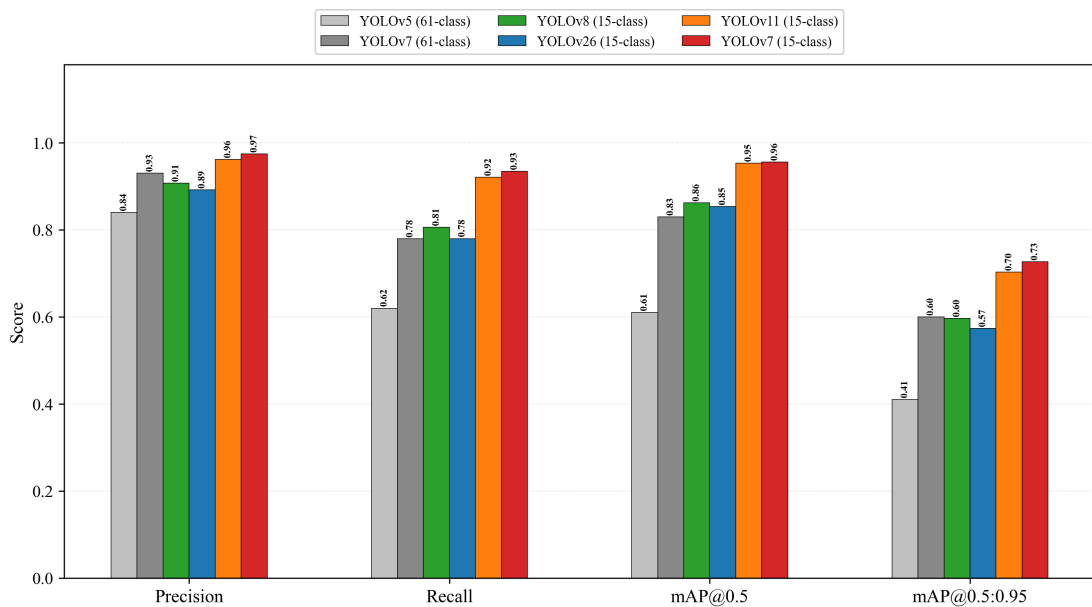


Figure 6-8 Grouped bar chart comparing all six detection configurations across Precision, Recall, mAP@0.5, and mAP@0.5:0.95

This section presents a unified view of all six detection configurations evaluated throughout the project, spanning two class taxonomies and five architectures. Table 6-4 and Figure 6-8 consolidate these results.

6.3.1 Object Detection Performance Evolution

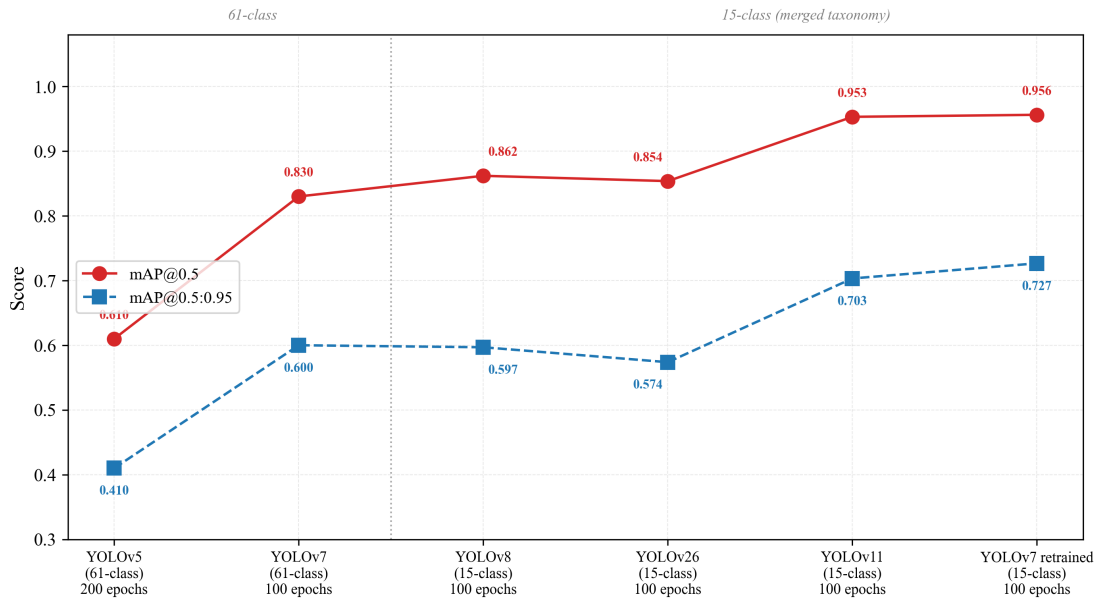


Figure 6-9 Performance evolution of object detection across all configurations showing mAP@0.5 and mAP@0.5:0.95

Figure 6-9 traces the evolution of detection performance across the project timeline. The trajectory shows two distinct phases of improvement: (1) moving from YOLOv5 to YOLOv7 on the 61-class taxonomy, and (2) consolidating the taxonomy to 15 classes and benchmarking multiple architectures. The retrained YOLOv7 on 15 classes achieves the highest performance, with mAP@0.5 improving from 0.61 (YOLOv5, 61-class) to 0.9562, a cumulative improvement of 56.8%.

6.3.2 Radar Visualization of Top Models

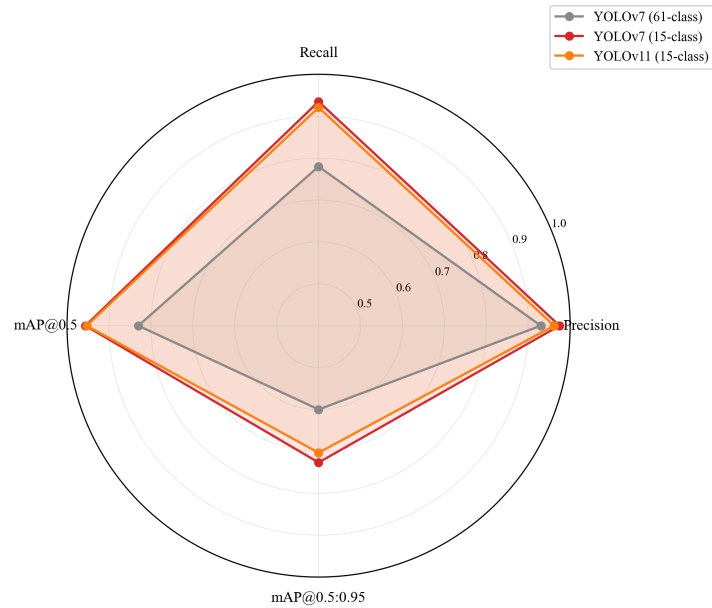


Figure 6-10 Radar chart comparing the three best detection configurations across four performance metrics

Figure 6-10 provides a radar chart comparing the three best-performing configurations. While YOLOv7 (61-class) shows a notable deficit in recall (0.78 vs. 0.93+), both 15-class models exhibit near-complete coverage of the performance space. The retrained YOLOv7 (15-class) marginally outperforms YOLOv11 across all axes, making it the model of choice for the final pipeline.

Overall, from all these findings, it is evident that YOLOv7 (retrained on the 15-class taxonomy) is the optimal object detection model for this project. This also goes to show that class taxonomy might matter more than the architecture at times.

6.3.3 Confusion Matrix for the retrained YOLOv7

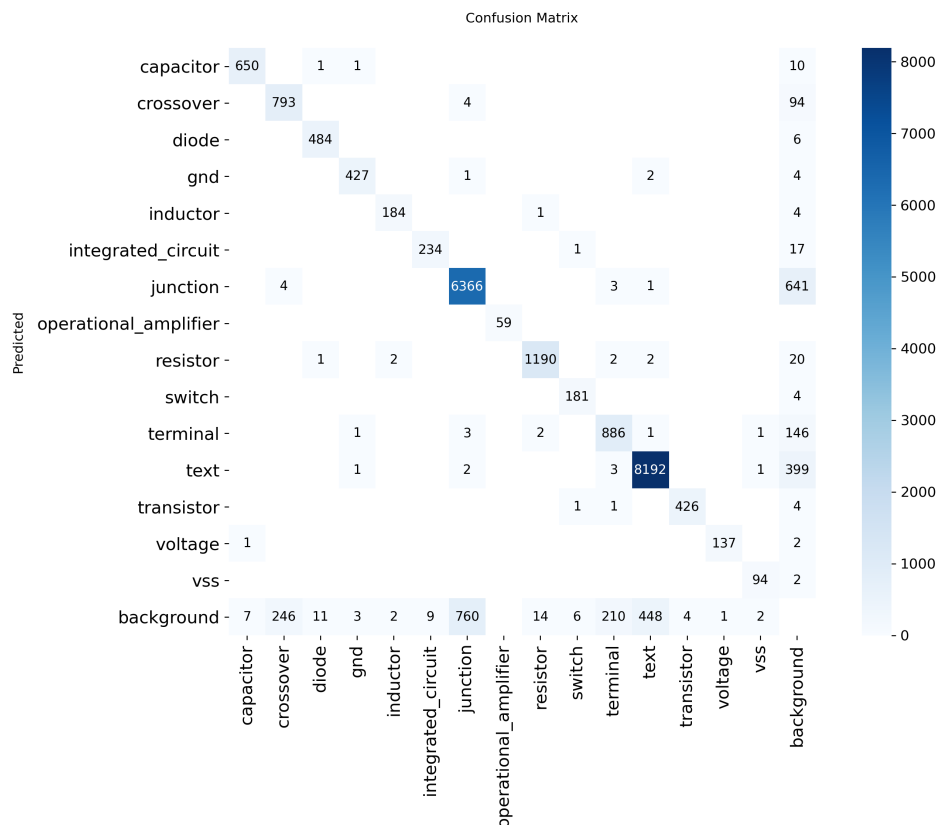


Figure 6-11 Confusion Matrix for the retrained YOLOv7

Figure 6-11 demonstrates the confusion matrix for the retrained YOLOv7 (15 classes). It has a darker main diagonal and very light off-diagonals, which means that the model exhibits strong class separability with minimal inter-class confusion.

6.4 Retrained YOLOv7 Demonstration

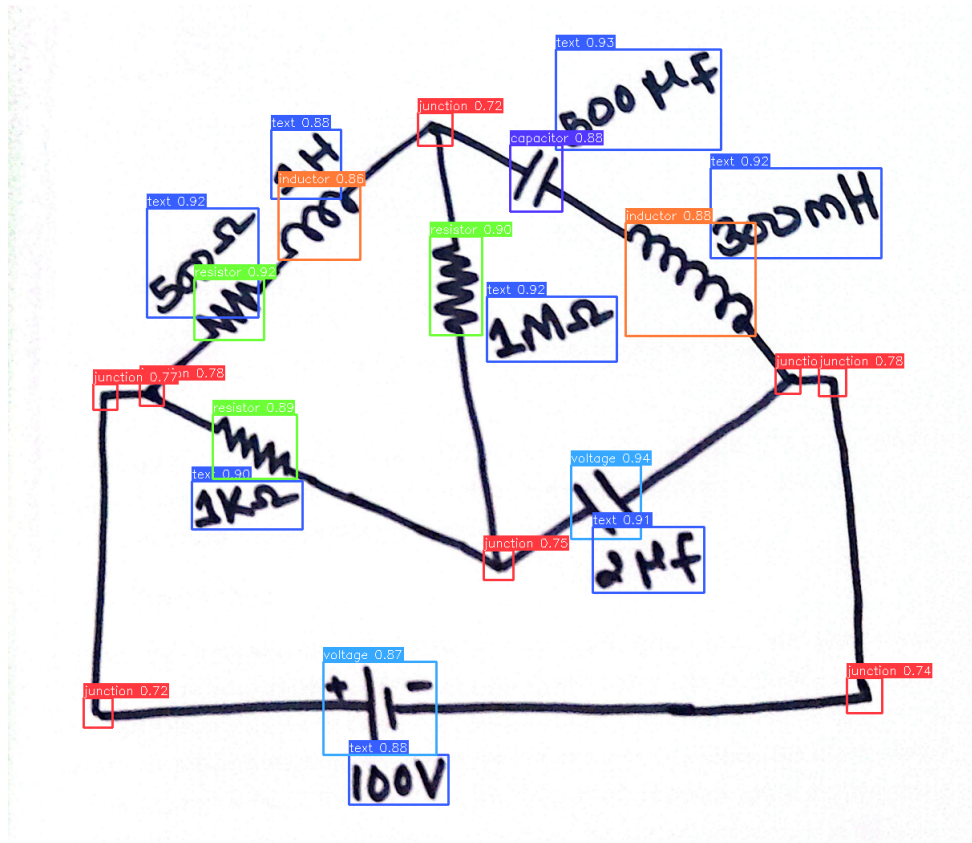


Figure 6-12 Retrained YOLOv7 Result (i)

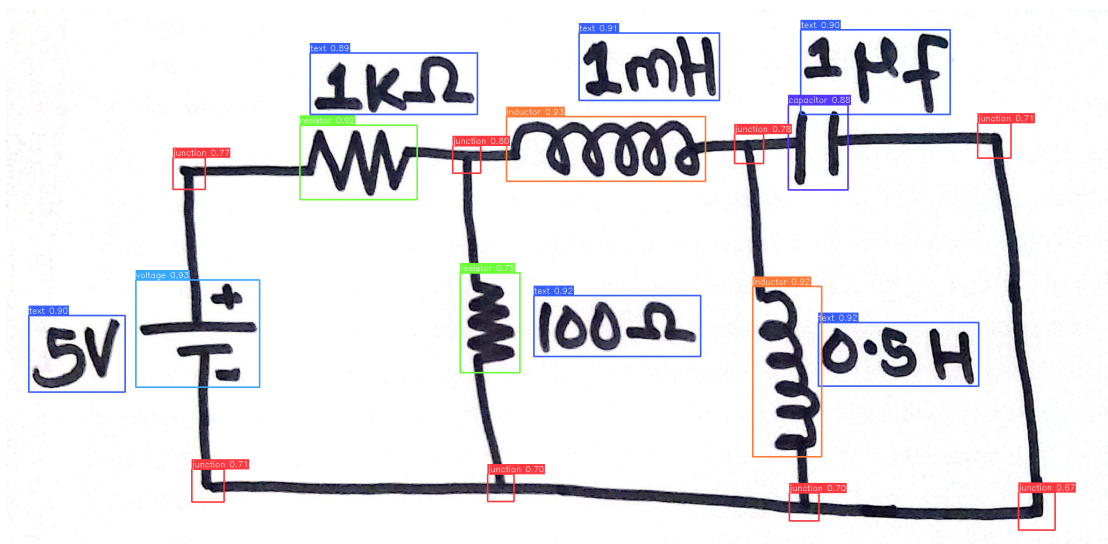


Figure 6-13 Retrained YOLOv7 Result (ii)

Figure 6-12 and Figure 6-13 show the results of the retrained YOLOv7 on two random test images. The model successfully detects and classifies all components in both images

with high confidence scores. These results visually confirm the quantitative metrics and demonstrate the model's effectiveness in real-world scenarios.

6.5 Custom OCR Results and Analysis

6.5.1 Custom OCR Training Loss Curve

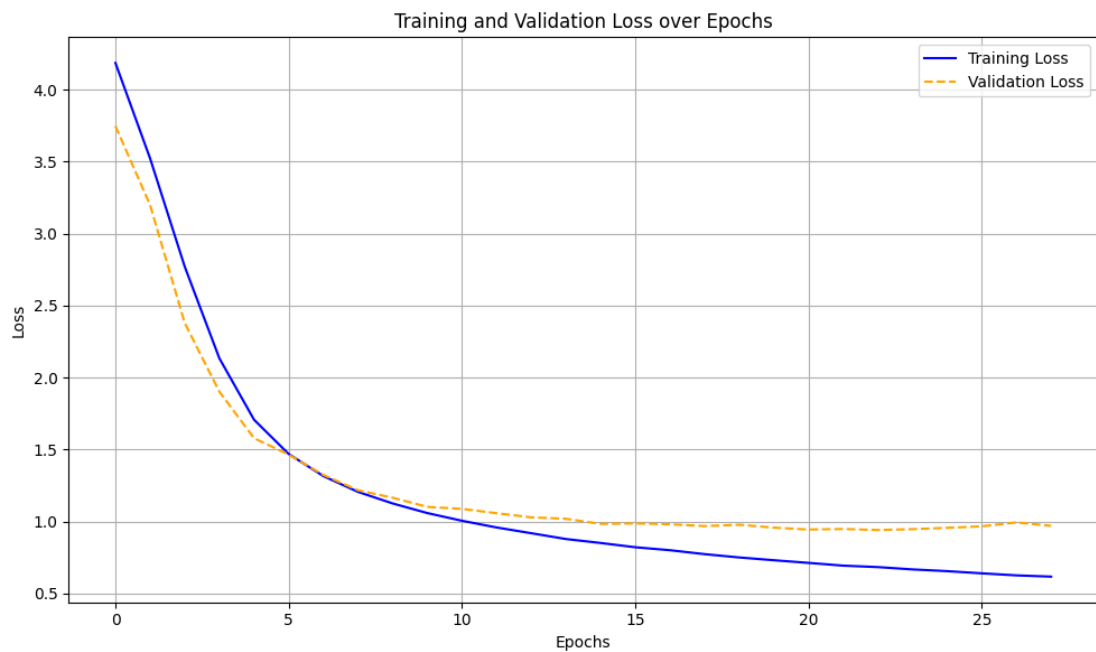


Figure 6-14 Training Loss vs Epochs (Custom OCR)

Figure 6-14 shows that the training loss decreases consistently throughout the 27 epochs, indicating the model is effectively memorizing the training data. There seems to be the slightest uptick in the validation loss, though, at 26 epoch indicating a hint of potential overfitting. The training was stopped at 27 epochs due to early stopping criteria being met.

6.5.2 Custom OCR Performance Analysis

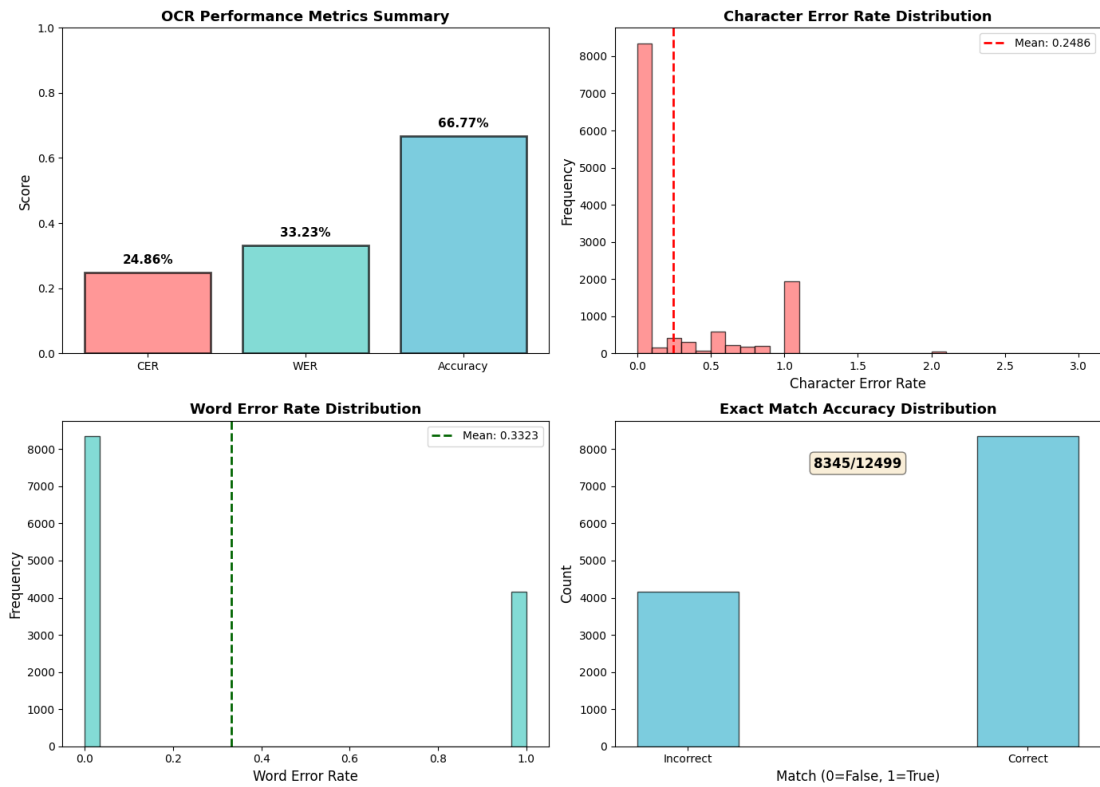


Figure 6-15 Custom OCR Performance Metrics and their Distributions

Figure 6-15 is a combination of images showing the performance metrics for the custom OCR model. The accuracy is 66.77%. Likewise, the CER of approximately 0.25 means that on average, 1 out of every 4 characters is predicted incorrectly. However, the histogram shows a notable bump at 1.0 (complete failure) confirming the presence of "garbage" inputs which can be fixed by the use of data cleaning techniques. WER is shown to be 0.33. Its distribution is inherently binary because the model either gets the full component value right (WER being 0) or gets it wrong (WER = 1).

The performance metrics indicate that while the custom OCR model has learned to recognize some component values, there is significant room for improvement. This sets the stage for TrOCR which is expected to have much better performance due to its pre-trained weights and more advanced architecture.

6.6 Comparative Analysis of OCR Models

Two distinct OCR architectures were evaluated for extracting component values (e.g., "10kΩ", "100μF") from detected text regions: the custom CRNN-based model trained from scratch, and a fine-tuned TrOCR-small model based on the Vision Transformer architecture as mentioned in previous sections.

6.6.1 Quantitative Comparison

Table 6-5 Performance comparison of OCR models

Metric	Custom CRNN	Fine-tuned TrOCR
Accuracy (%)	66.77	84.50
Character Error Rate (CER)	0.25	0.12
WER	0.33	0.18
Training Epochs	27	3
Batch Inference	Sequential	Parallel (GPU)
Precision (float)	float32	float16

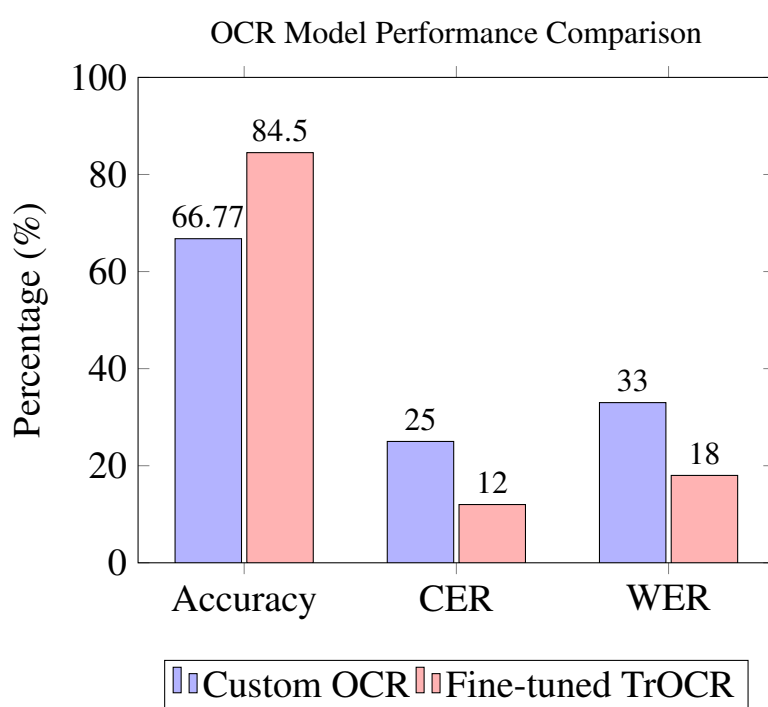


Figure 6-16 Performance metrics comparison, in percentage

Table 6-5 summarizes the performance of both OCR models, and Figure 6-16 provides a visual comparison. The fine-tuned TrOCR model outperforms the custom CRNN across all metrics despite requiring only 3 training epochs compared to 27 (tested on the test split of the CGHD dataset). The CER reduction from 0.25 to 0.12 indicates that TrOCR makes half as many character-level errors, directly improving the accuracy of extracted component values. It is noted that the custom CRNN's lower accuracy (66.77%) was partly attributed to data corruption during preprocessing. Regardless, the architectural advantages of the Transformer-based approach make TrOCR the superior choice for this pipeline. However, the inference speed of TrOCR has found to be considerably slower than the custom OCR model. This implies there is inherently a trade-off between accuracy and inference speed when choosing between the custom OCR and TrOCR

models for this project. Keeping this in mind, both models are retained in the codebase and can be switched based on user preference.

6.7 OCR Demonstration

Two randomly picked images from the dataset are shown below after passing through the detection (YOLOv7) and recognition (OCR) models.

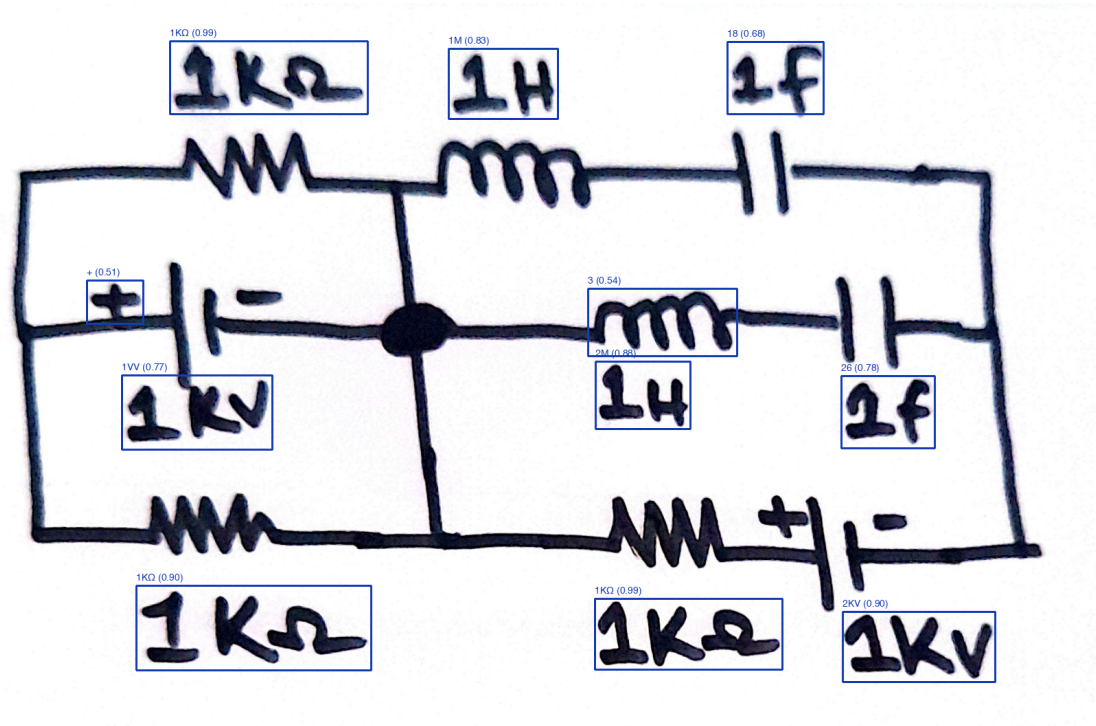


Figure 6-17 Text Detection and Recognition using Custom OCR(i)

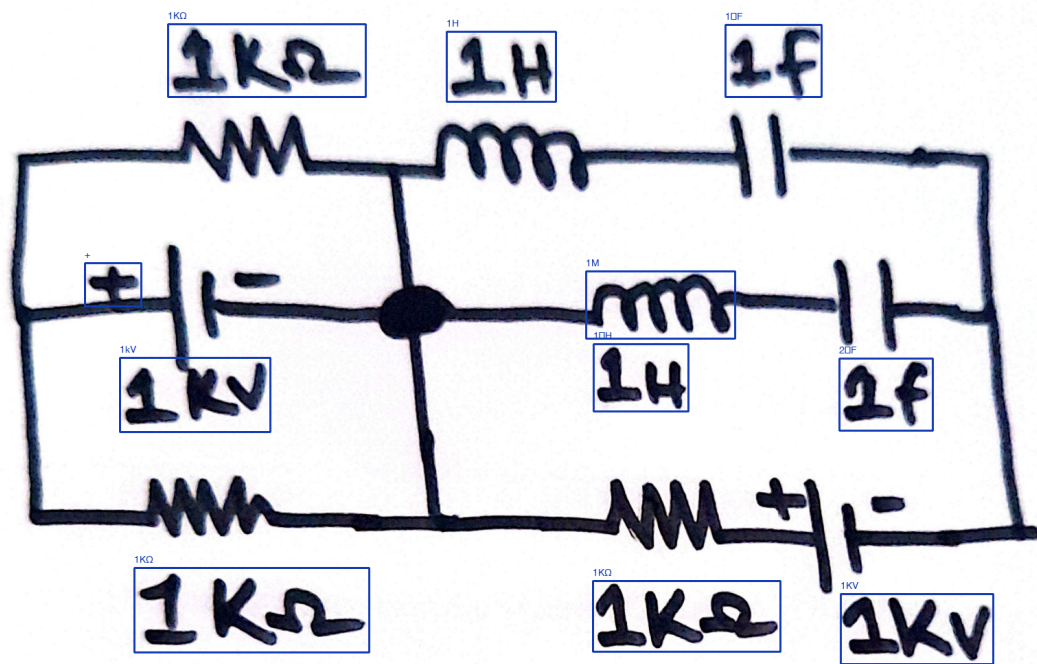


Figure 6-18 Text Detection and Recognition using TrOCR(i)

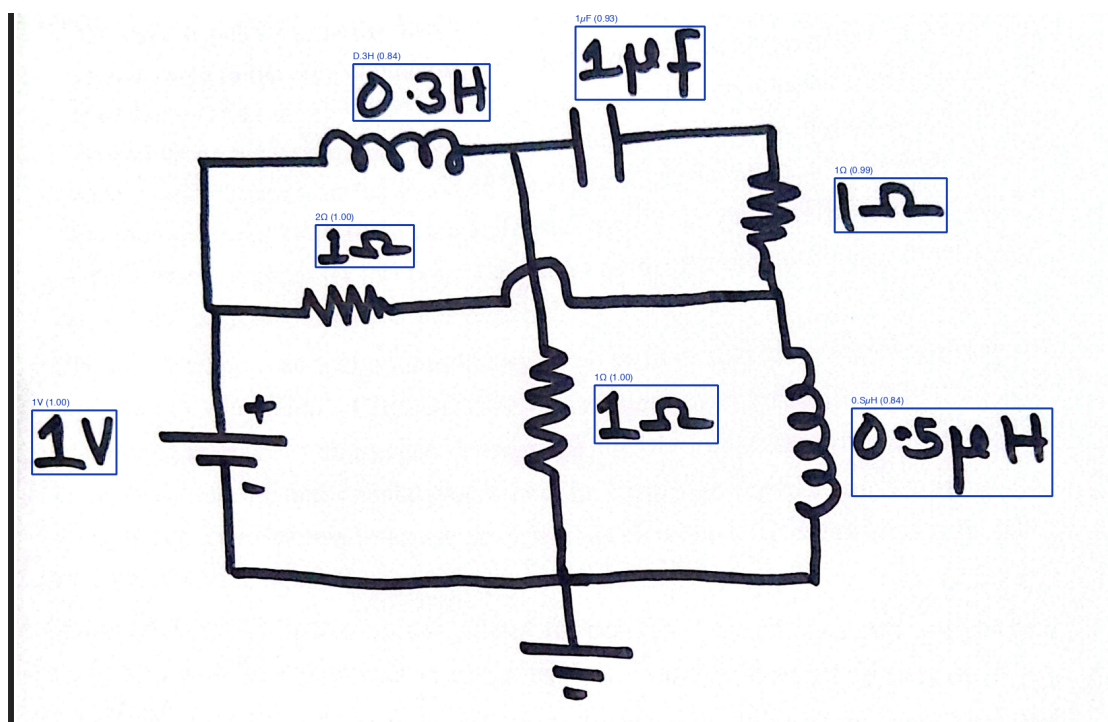


Figure 6-19 Text Detection and Recognition using Custom OCR(ii)

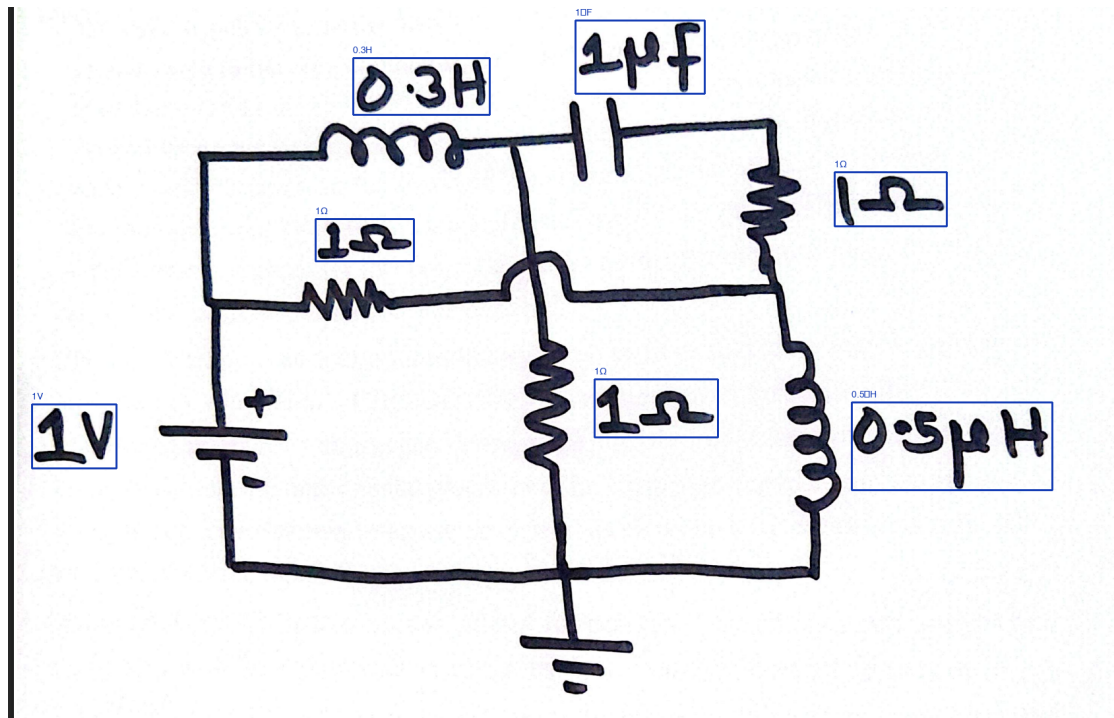


Figure 6-20 Text Detection and Recognition using TrOCR(ii)

Figure 6-17 and Figure 6-19 show the results of the custom OCR model on two random test images. The model successfully detects text regions and extracts component values, albeit with many errors. In contrast, Figure 6-18 and Figure 6-20 demonstrate the superior performance of the fine-tuned TrOCR model, which correctly recognizes most component values.

6.8 Junction-Based Wire Detection

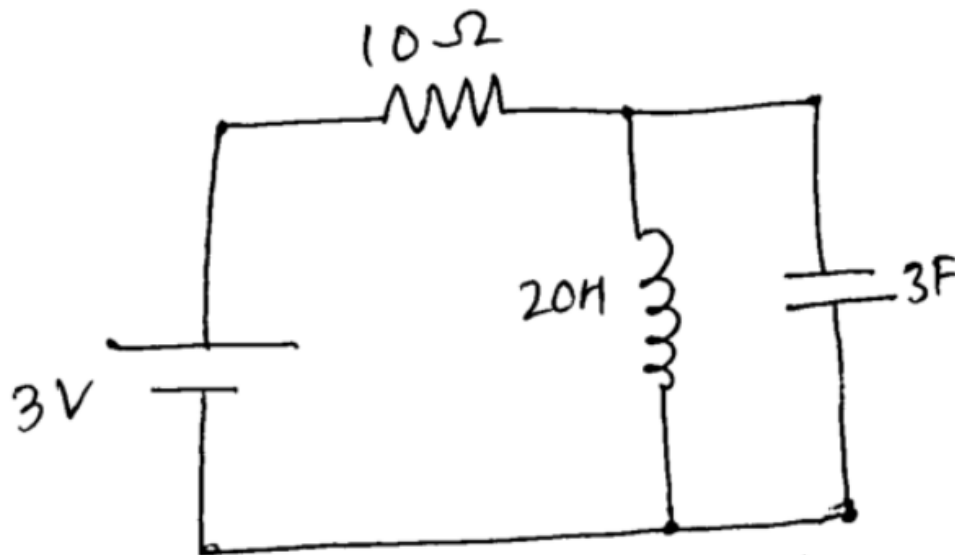


Figure 6-21 Otsu's Global Thresholding

This section presents the experimental results obtained from the junction-based wire detection pipeline applied to circuit diagram images. A high-resolution scanned circuit diagram is used as the input image. The original image dimensions are reported in terms of width and height in pixels to establish spatial scale consistency throughout the pipeline. To enhance structural features and suppress background noise, the input image is converted to grayscale. Otsu's global thresholding method is then applied to automatically compute an optimal threshold value that separates foreground circuit elements from the background. The resulting binary image preserves wires, junctions, and components while eliminating illumination variations as shown in Figure 6-21.

Morphological skeletonization is performed on the binary image to reduce all lines to a single-pixel width while maintaining connectivity. This process simplifies the subsequent analysis of wire paths and junctions. The skeletonized image is then overlaid with bounding boxes obtained from the object detection stage (i.e., YOLOv7) to identify potential endpoints where wires intersect component boundaries. Detected endpoints are marked on the skeleton, and their spatial coordinates are recorded for further processing.

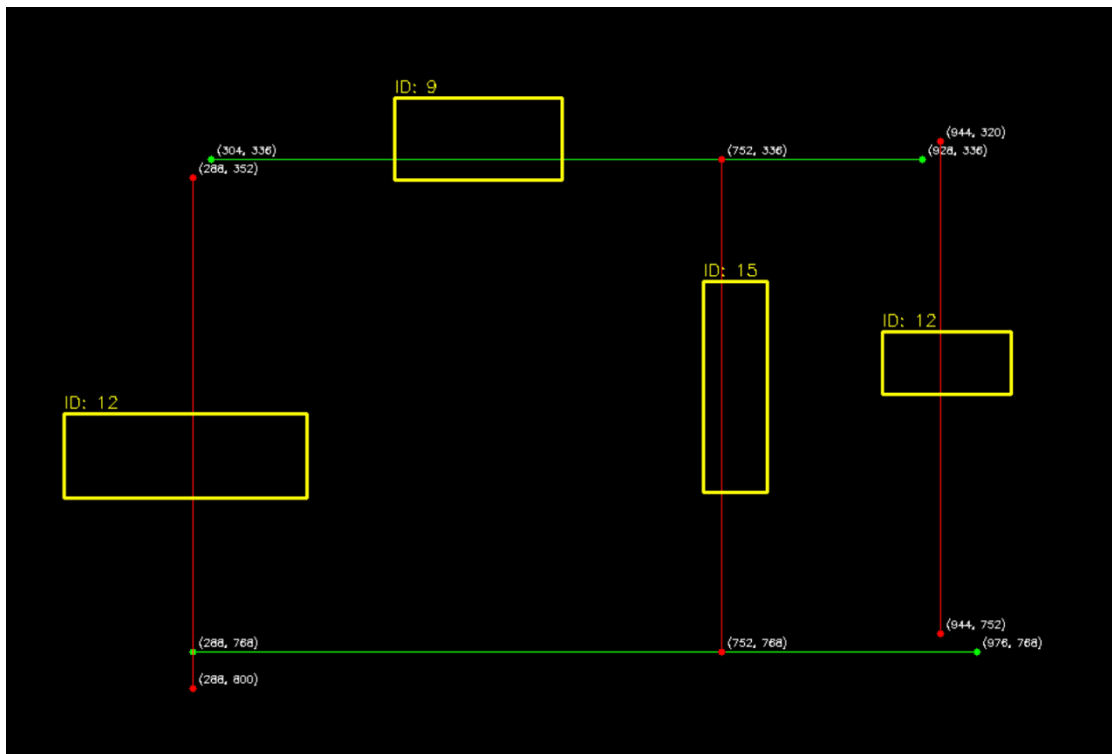


Figure 6-22 Bounding Box Overlay

Figure 6-22 shows the skeletonized image with bounding boxes overlaid. Intersection points between wires and component bounding box edges are computed to identify electrical connectivity. The total number of intersection points is reported along with a directional breakdown for horizontal and vertical wires intersecting specific component edges. These intersections form the basis for reconstructing circuit connectivity graphs and CircuitJS-compatible text files.

However, this junction-based method has a very important shortcoming: it fails to handle diagonal lines. This is because the bounding box intersection method relies on the assumption that wires will intersect component boundaries along horizontal or vertical axes. Diagonal wires may not intersect the bounding box edges in a predictable manner, leading to missed connections and incomplete graph construction. Therefore, this method is rejected in favor of a more robust path-finding approach that can handle arbitrary wire orientations and junction configurations.

6.9 Line Detection via Path Finding Visualization

6.9.1 Skeletonized Image with Detected Endpoints

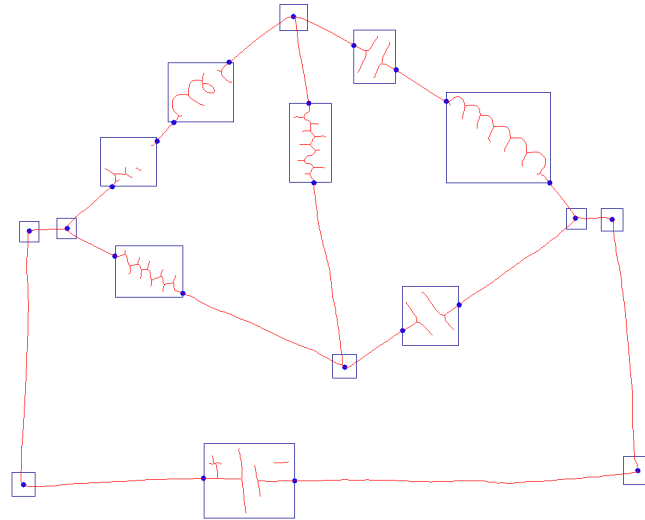


Figure 6-23 Skeletonized image with detected endpoints

Figure 6-23 displays the skeletonized image with all detected endpoints overlaid. The skeleton representation captures the one-pixel-wide structure of the circuit lines after morphological processing. Blue dots indicate the detected endpoints, which were identified using the bounding box intersection method for line segments and center-snapping for junction points. The blue rectangles show the original YOLO bounding boxes used for endpoint localization. This visualization confirms that the endpoint detection algorithm successfully identifies connection points at the boundaries of detected objects.

6.9.2 Adjacency Graph Visualization

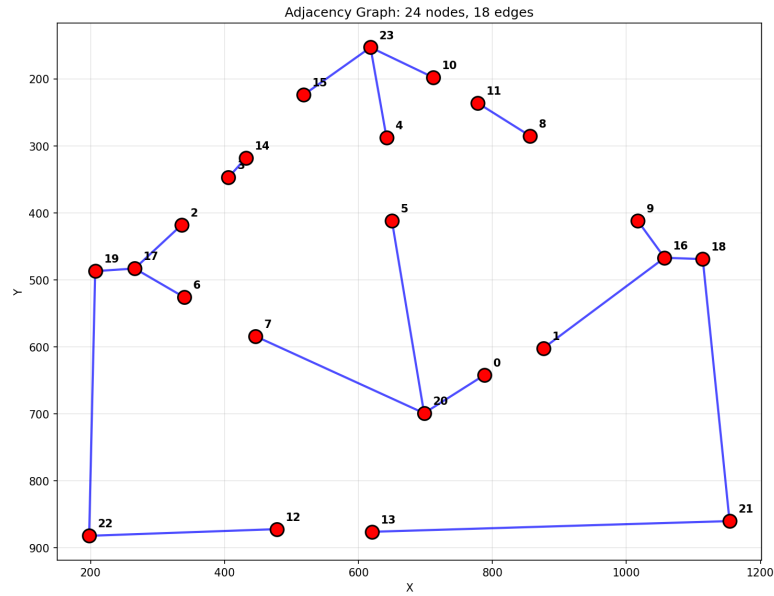


Figure 6-24 Adjacency graph constructed using Multi-Head BFS algorithm

Figure 6-24 presents the final adjacency graph constructed using the Multi-Head BFS algorithm. Each node corresponds to a detected endpoint from the previous stage. Blue lines connecting the nodes represent edges in the graph, indicating skeleton connectivity between endpoints. The graph has been processed to remove self-edges (connections between endpoints belonging to the same detected object), ensuring that only meaningful inter-component connections are preserved. This graph structure enables downstream applications such as circuit topology analysis and netlist extraction.

6.10 Combined Overlay of YOLOv7, TrOCR, and Line Detection Results

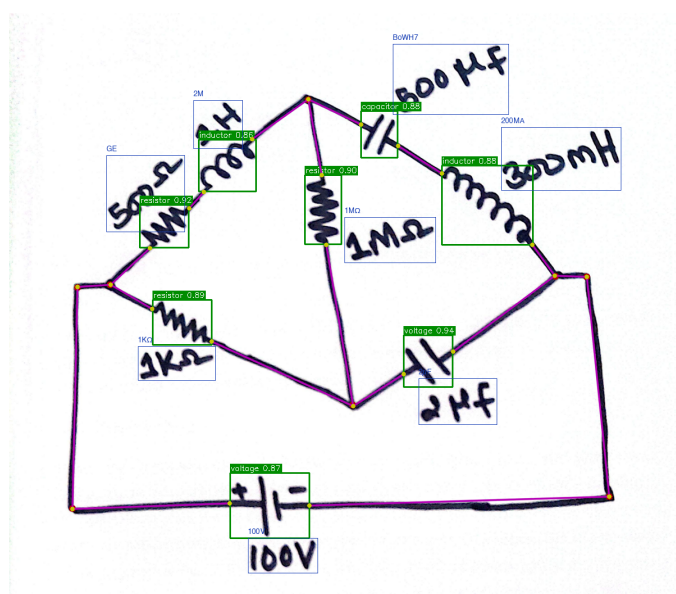


Figure 6-25 Combined overlay of YOLOv7, TrOCR, and line detection results

Figure 6-25 shows a comprehensive overlay of the results from the object detection (YOLOv7), text recognition (TrOCR), and line detection (adjacency graph) stages. The original circuit diagram is displayed in the background, with YOLOv7 bounding boxes highlighted in green, TrOCR-extracted component values annotated in blue text, and detected connections between components represented as red lines. This integrated visualization confirms that the pipeline successfully identifies components, extracts their values, and reconstructs connectivity, providing a holistic view of the circuit diagram interpretation process.

6.11 Frontend Visualization

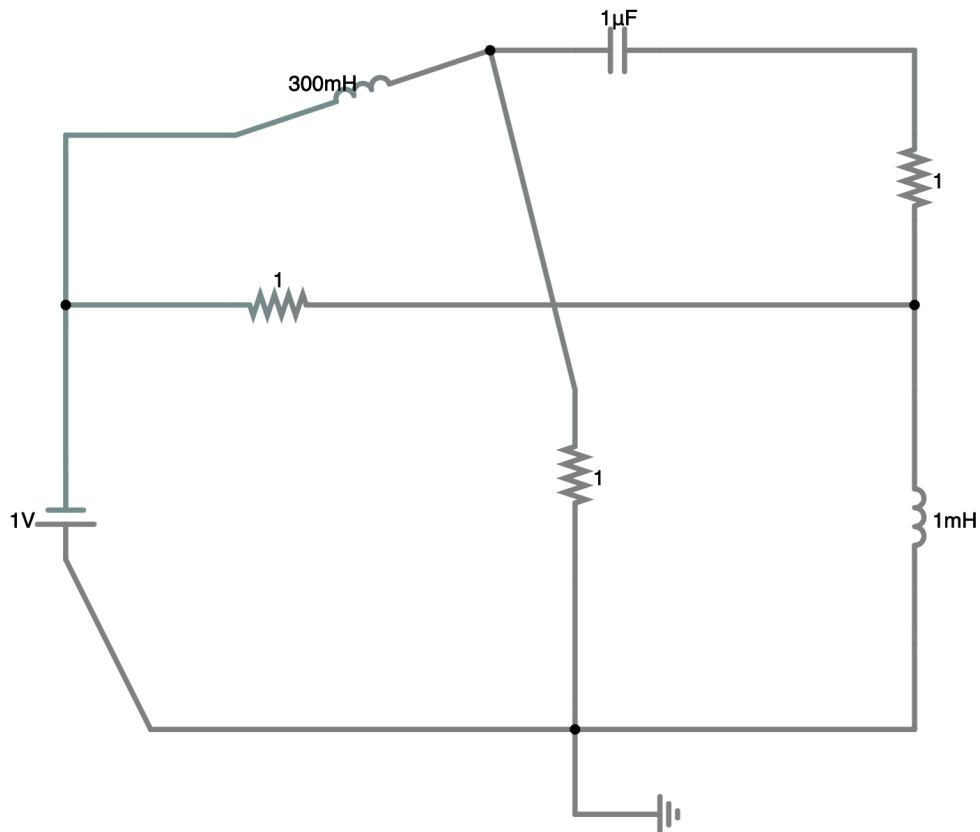


Figure 6-26 Digitally Drawn Circuit (Color Inverted)

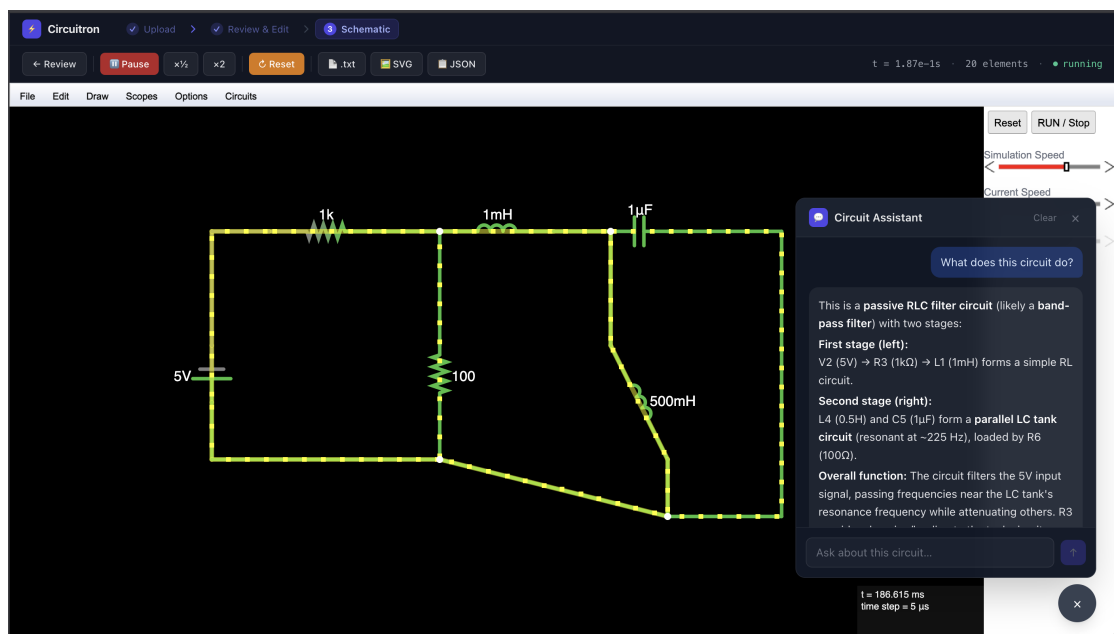


Figure 6-27 Fullscreen Capture of the Frontend

Figure 6-26 shows an editable, digital circuit auto-generated by the system. This is the output before any manual intervention. Figure 6-27 shows the fullscreen capture of the frontend with a sample circuit as well as the Q&A section using DeepSeek v3.1.

The frontend of the Circuitron web application provides an interactive environment for circuit design, simulation, and analysis. It enables users to construct electrical circuits using a drag-and-drop schematic editor with grid-based alignment, snap-to-grid functionality, and real-time wire routing. Components such as passive elements, active devices, and sources can be configured through a properties panel that allows precise control over electrical parameters and placement.

The system integrates simulation controls for transient analysis, allowing users to define time parameters, execute simulations, and visualize results through dynamically rendered voltage-time plots. Measurement probes support real-time signal monitoring with automated computation of key metrics, including maximum, minimum, peak-to-peak range, and Root Mean Square (RMS) values. Additionally, the frontend supports exporting simulation data and visualizations in standard formats, enabling seamless documentation and further offline analysis.

7 FUTURE ENHANCEMENTS

The CIRCUITRON system, while fully functional in its current state, has numerous opportunities for future enhancements. One of the objectives of any research-based project is to encourage the scientific community to conduct further studies and experiment on the working principle of the project to either verify the methodology used, or potentially come up with alternate techniques to solve the same problem. In any case, following are the likely future enhancements that can be implemented in this project:

7.1 Improvement in the OCR accuracy

The system uses two OCR models, at the moment, however the accuracy of both of them (highest of them being around 85%) has room for improvement. Also, the issue with TrOCR (which is the better of the two) is that it is too slow on account of its large size. The custom OCR pipeline is much faster, but has lower accuracy. Future work can be done to either improve the accuracy of the custom OCR model through rigorous training on different datasets, or to somehow improve TrOCR's speed.

7.2 Increment in the number of supported components

The system currently supports a limited number of components (only 12 after consolidation, minus crossover, junction, and text classes). Pre-consolidation, the system supported 59 components, but the mAP was too low for satisfactory object detection. These 12 components are the most commonly used ones for basic circuit design and simulation, however, there are many more components that can be added to the system to make it more versatile and useful for a wider range of applications. Future work can be done to increase the number of supported components, which would require training the object detection model on a larger and more diverse dataset.

7.3 Authentication and Database Functionalities

The system, in its current form, doesn't allow the user to save their work or to access it later. Moreover, there is no authentication mechanism in place to allow users to have personalized accounts and potentially collaborate on the same projects. This also involves hosting the system online. Future enhancements can include the integration of a database to store user data and circuit designs, as well as offer a library of pre-designed circuits for beginners to play around with.

7.4 Improvement in the frontend and general user experience

While the system is capable of handling diagonal wires and components, when said components are rendered in the frontend, they aren't rendered in the same diagonal

manner. This requires the user to step in and manually adjust the orientation of the component. The issue of scale, between the wires and the components also persists, occasionally requiring manual resizing. Moreover, the frontend of the system involves a lot of user interactions from dragging and resizing, to general drawing and editing of the schematic, which can be made smoother in the future.

8 CONCLUSION

The objectives of recognizing a hand-drawn circuit diagram, transforming it into an editable schematic having simulation capabilities, and analyzing said circuit with LLM-powered Q&A have been successfully achieved. Functional pipelines for each aspect of the project have been designed and implemented effectively. For component detection, YOLOv7 has achieved a mAP@0.5 of 95% and for text recognition, TrOCR has achieved accuracy of 84.5%. The wire detection algorithm leverages junction-based geometric inference with adaptive collinear merging, enabling robust extraction of circuit connectivity. The system processes hand-drawn circuit images and extracts components with their values and connections, effectively digitizing a circuit. The project has successfully addressed a significant gap in educational circuit analysis tools by making circuit simulation easier than ever.

Apart from technical performance, the project also demonstrates successful full-stack integration with a React.js frontend, a FastAPI backend, and an integration with CircuitJS for simulation of the circuit. This modular architecture ensures ease of maintenance and potential extensibility, establishing a solid foundation for future enhancements.

This project also serves as a vehicle for research and experimentation for its authors. The study into different versions of YOLO for component detection, different OCR models for text recognition, as well as multiple ways to solve wire connectivity issues has not only brought this project as close to the best as it could have been, but also infused the authors with the knowledge and technical know-how regarding a variety of topics and cutting-edge scientific research. Circuitron represents a meaningful step towards democratizing circuit education through the elimination of friction between ideation and analysis. It establishes a replicable and of course, improvable, design pattern for the broader electronics education community. Its potential to improve student engagement while reducing cognitive load makes it a valuable contribution to electrical and electronics education, where traditional circuit simulation tools are expensive or complex.

9 APPENDICES

Appendix A: Project Budget

The project has been completed with a total budget of NPR 41,160. This budget covers various aspects of the project, including computation resources, cloud services, IEEE membership, and miscellaneous expenses.

Table 9-1 Project Budget Breakdown

Category	Details	Estimated Cost (NPR)
Computation Requirements	Google Colab Pro (\$9.99 per month), NVIDIA T4 GPU (\$0.35 per hour), Lightning AI Credits	24,000
IEEE Xplore Membership	Annual subscription for accessing research articles, journals	2,160
Miscellaneous Costs	Report printing, binding, API, etc.	15,000
Total		41,160

Appendix B: Gantt Chart

Table 9-2 Project Timeline (Gantt Chart)

PROCESS	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Research											
Implementation											
Testing											
Documentation											
Final Submission											

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